

Algebra Foundations

Shel Scholars Competitive Math Academy

Week 0

Introduction

This handout covers the algebraic foundations that we will assume throughout the AMC 10 Get 100 Camp.

Most of the ideas in Week 0 should not be completely new. You should already have seen things like order of operations, fractions, negative numbers, basic equations, rates, unit conversions, and percentages. Our goal is to make sure that these skills are reliable enough that they do not become the reason you miss an AMC 10 problem.

That distinction matters. Many early AMC 10 problems do not require advanced mathematics. Instead, they take familiar ideas and present them in an unfamiliar form: a complicated-looking expression, a word problem whose equation is hidden in the language, several quantities measured in different units, or successive percentage changes with different bases.

So Week 0 has two purposes:

1. **Review the algebra we will assume you know.** If something here feels unfamiliar, this is the time to identify it and learn it before the main camp begins.
2. **Learn to recognize these basic ideas when they are disguised inside AMC problems.** The arithmetic may be elementary even when the problem does not look elementary.

You should aim to finish Week 0 able to carry out the computations in this handout accurately and, more importantly, to recognize which basic tool a problem is asking you to use.

Main idea/Fact (What Week 0 is for)

Week 0 is prerequisite review, not the main algebra curriculum of the camp.

The goal is not merely to know these ideas. The goal is to be able to use them quickly, accurately, and without them getting in the way of the harder mathematics that comes later.

Simple Computational Problems

The Problem

In the AMC 10 you are often given problems like the following:

Example 1 (2024 AMC 10A, Problem 1)

What is the value of $9901 \cdot 101 - 99 \cdot 10101$?

(A) 2 (B) 20 (C) 21 (D) 200 (E) 2020

You can multiply the things together and get the correct solution or you can realize that the things inside the expression are pretty similar.

Namely, both 99 and 101 are close to 100, they are 1 off. Also 9901 and 10101 are both close to $100^2 = 10000$.

Now the main idea is if we set $k = 100$ and write things in terms of k we'll get something easier to deal with. Namely we have:

$$\begin{aligned} 99 &= k - 1, & 101 &= k + 1 \\ 9901 &= k^2 - k + 1, & 10101 &= k^2 + k + 1 \end{aligned}$$

So after we plug that in, we'll get the expression is:

$$\begin{aligned} 9901 \cdot 101 - 99 \cdot 10101 &= \\ (k^2 - k + 1)(k + 1) - (k - 1)(k^2 + k + 1) &= \\ (k^3 + 1) - (k^3 - 1) &= 2 \end{aligned}$$

And ergo the correct answer is 2.

Sometimes there are tricks like in this one that make the computation easier and sometimes you just have to make sure you're not making an arithmetic mistake somewhere.

This is the height of skill when it comes to solving problems that are simply computational. However we're going to start with the basics in this lecture and move our way up.

Introduction

The point of this lecture is not to become a calculator. The point is to become reliable at evaluating expressions that look messy but only use simple rules.

A lot of early AMC 10 problems are computational. This means that the main difficulty is not a deep idea, but clean execution: knowing what to simplify first, keeping signs correct, and not losing part of the expression.

Main idea/Fact (Main habit for computational problems)

After each operation, rewrite the whole expression.

Example 2

Evaluate

$$18 - 2(3 + 4).$$

Solution. We first simplify the grouped part:

$$18 - 2(3 + 4) = 18 - 2(7).$$

Now multiply:

$$18 - 2(7) = 18 - 14.$$

Finally subtract:

$$18 - 14 = 4.$$

Therefore the value is

$$\boxed{4}.$$

Pitfall

If you try to do all the arithmetic in your head, the numbers may be easy but the bookkeeping becomes dangerous. Rewriting the expression after every step makes sign mistakes much easier to catch.

Expressions are instructions

An expression tells us what operations to do and in what order. If we change the order, we may change the answer.

Example 3

Compare

$$(8 - 3) \cdot 2$$

and

$$8 - 3 \cdot 2.$$

Solution. In the first expression, the parentheses tell us to subtract first:

$$(8 - 3) \cdot 2 = 5 \cdot 2 = 10.$$

In the second expression, multiplication comes before subtraction:

$$8 - 3 \cdot 2 = 8 - 6 = 2.$$

So the two expressions have different values:

$$(8 - 3) \cdot 2 = 10, \quad 8 - 3 \cdot 2 = 2.$$

Main idea/Fact (Order of operations)

We evaluate in the following order:

Grouping $\longrightarrow$ Exponents $\longrightarrow$ Multiplication and Division $\longrightarrow$ Addition and Subtraction.

Multiplication and division are tied, so we go left to right. Addition and subtraction are tied, so we go left to right.

Example 4

Evaluate

$$5 + 3 \cdot 2^3.$$

Solution. First evaluate the exponent:

$$5 + 3 \cdot 2^3 = 5 + 3 \cdot 8.$$

Now multiply:

$$5 + 3 \cdot 8 = 5 + 24.$$

Now add:

$$5 + 24 = 29.$$

Therefore the value is

$$\boxed{29}.$$

Multiplication and division are tied

Multiplication and division have the same priority. We do not always multiply before dividing. We move from left to right.

Example 5

Evaluate

$$24 \div 3 \cdot 2.$$

Solution. Since multiplication and division are tied, we go left to right. The division comes first:

$$24 \div 3 \cdot 2 = 8 \cdot 2.$$

Now multiply:

$$8 \cdot 2 = 16.$$

Therefore

$$24 \div 3 \cdot 2 = \boxed{16}.$$

Pitfall

A common mistake is to multiply $3 \cdot 2$ first:

$$24 \div (3 \cdot 2) = 24 \div 6 = 4.$$

But this is not the original expression. The original expression has no parentheses around $3 \cdot 2$.

Addition and subtraction are tied

Addition and subtraction also have the same priority. We do not always add before subtracting. We move from left to right.

Example 6

Evaluate

$$10 - 4 + 3 - 8.$$

Solution. Go from left to right:

$$10 - 4 + 3 - 8 = 6 + 3 - 8.$$

Now continue:

$$6 + 3 - 8 = 9 - 8.$$

Finally:

$$9 - 8 = 1.$$

Therefore the value is

$$\boxed{1}.$$

Grouping symbols come first

Parentheses are grouping symbols, but they are not the only grouping symbols.

Main idea/Fact

Grouping symbols.

The following all create grouped expressions:

$$(), \quad [], \quad \{ \}, \quad \frac{\square}{\square}, \quad \sqrt{}.$$

A fraction bar groups the whole numerator and the whole denominator.

Example 7

Evaluate

$$\frac{5 + 7}{3 - 1}.$$

Solution. The numerator and denominator are separate grouped expressions. First simplify the top and the bottom:

$$\frac{5 + 7}{3 - 1} = \frac{12}{2}.$$

Now divide:

$$\frac{12}{2} = 6.$$

Therefore the value is

$$\boxed{6}.$$

Example 8

Evaluate

$$2[3 + (4 - 1)].$$

Solution. Start with the innermost parentheses:

$$2[3 + (4 - 1)] = 2[3 + 3].$$

Now simplify inside the brackets:

$$2[3 + 3] = 2[6].$$

Now multiply:

$$2[6] = 12.$$

Therefore the value is

$$\boxed{12}.$$

Pitfall

When grouping symbols are nested, simplify from the inside out.

Nested fractions

When a fraction contains another fraction inside it, start from the deepest fraction and work outward.

Example 9 (2022 AMC 10A, Problem 1)

What is the value of

$$3 + \frac{1}{3 + \frac{1}{3 + \frac{1}{3}}}?$$

Solution. We work from the bottom up.

First simplify the deepest denominator:

$$3 + \frac{1}{3} = \frac{9}{3} + \frac{1}{3} = \frac{10}{3}.$$

So the expression becomes

$$3 + \frac{1}{3 + \frac{1}{10/3}}.$$

Now

$$\frac{1}{10/3} = \frac{3}{10}.$$

Thus

$$3 + \frac{1}{3 + \frac{1}{10/3}} = 3 + \frac{1}{3 + \frac{3}{10}}.$$

Simplify the denominator:

$$3 + \frac{3}{10} = \frac{30}{10} + \frac{3}{10} = \frac{33}{10}.$$

So

$$3 + \frac{1}{3 + \frac{3}{10}} = 3 + \frac{1}{33/10} = 3 + \frac{10}{33}.$$

Finally,

$$3 + \frac{10}{33} = \frac{99}{33} + \frac{10}{33} = \boxed{\frac{109}{33}}.$$

Main idea/Fact

Nested fraction habit.

If the expression is a fraction inside a fraction, work from the deepest layer outward.

Exponents and negative signs

An exponent applies only to its base. Parentheses decide whether the negative sign is part of the base.

Example 10

Compare

$$(-3)^2$$

and

$$-3^2.$$

Solution. In $(-3)^2$, the base is -3 , so

$$(-3)^2 = (-3)(-3) = 9.$$

In -3^2 , the exponent applies only to 3, and the negative sign stays outside:

$$-3^2 = -(3^2) = -9.$$

Therefore

$$(-3)^2 = 9, \quad -3^2 = -9.$$

Pitfall

Parentheses decide the base:

$$(-a)^2 = a^2, \quad -a^2 = -(a^2).$$

These are not the same expression.

Absolute value and square roots

The absolute value of a number is its distance from 0. Therefore it is always nonnegative.

Example 11

Evaluate

$$|-5|, \quad |7|, \quad |0|.$$

Solution. The number -5 is distance 5 from 0, so

$$|-5| = 5.$$

The number 7 is distance 7 from 0, so

$$|7| = 7.$$

The number 0 is distance 0 from 0, so

$$|0| = 0.$$

Main idea/Fact

Square root of a square.

For every real number x ,

$$\sqrt{x^2} = |x|.$$

Example 12

Evaluate

$$\sqrt{(-7)^2}.$$

Solution. We first compute inside:

$$(-7)^2 = 49.$$

So

$$\sqrt{(-7)^2} = \sqrt{49} = 7.$$

This agrees with the rule

$$\sqrt{x^2} = |x|,$$

because

$$|-7| = 7.$$

Absolute value with variables

When an absolute value contains a variable expression, first decide whether the expression inside is positive or negative.

Example 13 (2022 AMC 10A, Problem 6)

For $a < 0$, simplify

$$\left| a - 2 - \sqrt{(a-1)^2} \right|.$$

Solution. Since $a < 0$, we know

$$a - 1 < 0.$$

Now use

$$\sqrt{x^2} = |x|.$$

This gives

$$\sqrt{(a-1)^2} = |a-1|.$$

Because $a - 1 < 0$, we have

$$|a-1| = -(a-1) = 1-a.$$

Substitute this into the original expression:

$$\left| a - 2 - \sqrt{(a-1)^2} \right| = |a - 2 - (1-a)|.$$

Simplify inside:

$$a - 2 - (1 - a) = a - 2 - 1 + a = 2a - 3.$$

Therefore the expression becomes

$$|2a - 3|.$$

Since $a < 0$, we have

$$2a - 3 < 0.$$

So

$$|2a - 3| = -(2a - 3) = 3 - 2a.$$

Therefore the simplified expression is

$$\boxed{3 - 2a}.$$

Pitfall

Do not write

$$\sqrt{(a - 1)^2} = a - 1$$

unless you know $a - 1 \geq 0$. In this problem $a < 0$, so $a - 1 < 0$, and the correct simplification is

$$\sqrt{(a - 1)^2} = |a - 1| = 1 - a.$$

Defined operators

Sometimes a problem defines a new symbol. The symbol is just a rule. Every time the symbol appears, replace it using the rule.

Example 14

Suppose

$$x \diamond y = |x - y|.$$

Evaluate

$$2 \diamond 5.$$

Solution. Use the definition:

$$2 \diamond 5 = |2 - 5|.$$

Now simplify:

$$|2 - 5| = |-3| = 3.$$

Therefore

$$2 \diamond 5 = \boxed{3}.$$

Example 15

Suppose

$$x \diamond y = |x - y|.$$

Evaluate

$$1 \diamond (2 \diamond 3).$$

Solution. Start inside the parentheses:

$$2 \diamond 3 = |2 - 3| = |-1| = 1.$$

So

$$1 \diamond (2 \diamond 3) = 1 \diamond 1.$$

Now use the definition again:

$$1 \diamond 1 = |1 - 1| = 0.$$

Therefore

$$1 \diamond (2 \diamond 3) = \boxed{0}.$$

Defined operators on the AMC

Example 16 (2022 AMC 10B, Problem 1)

Define

$$x \diamond y = |x - y|$$

for all real numbers x and y . What is the value of

$$(1 \diamond (2 \diamond 3)) - ((1 \diamond 2) \diamond 3)?$$

Solution. Compute the left group first.

We have

$$2 \diamond 3 = |2 - 3| = |-1| = 1.$$

Therefore

$$1 \diamond (2 \diamond 3) = 1 \diamond 1 = |1 - 1| = 0.$$

Now compute the right group.

We have

$$1 \diamond 2 = |1 - 2| = |-1| = 1.$$

Therefore

$$(1 \diamond 2) \diamond 3 = 1 \diamond 3 = |1 - 3| = |-2| = 2.$$

Now substitute both values:

$$(1 \diamond (2 \diamond 3)) - ((1 \diamond 2) \diamond 3) = 0 - 2 = \boxed{-2}.$$

Pitfall

Defined operators do not have to behave like normal addition or multiplication. In this example,

$$1 \diamond (2 \diamond 3) \neq (1 \diamond 2) \diamond 3.$$

So the parentheses must be followed exactly.

Factorials

A factorial means multiplying down to 1.

Definition

Factorial.

For $n \geq 1$,

$$n! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1.$$

Also,

$$0! = 1.$$

Example 17

Compute

$$5!.$$

Solution. By definition,

$$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1.$$

Therefore

$$5! = 120.$$

Example 18

Simplify

$$\frac{10!}{8!}.$$

Solution. Expand only until the $8!$ appears:

$$10! = 10 \cdot 9 \cdot 8!.$$

Therefore

$$\frac{10!}{8!} = \frac{10 \cdot 9 \cdot 8!}{8!} = 10 \cdot 9 = 90.$$

Main idea/Fact

Factorial cancellation habit.

When a factorial fraction appears, expand only as far as needed for cancellation.

Factorials on the AMC

Example 19 (2016 AMC 10A, Problem 1)

What is the value of

$$\frac{11! - 10!}{9!}?$$

Solution. Factor $10!$ out of the numerator:

$$11! - 10! = 11 \cdot 10! - 10! = (11 - 1)10! = 10 \cdot 10!.$$

Now rewrite $10!$ as $10 \cdot 9!$:

$$10 \cdot 10! = 10 \cdot 10 \cdot 9!.$$

Therefore

$$\frac{11! - 10!}{9!} = \frac{10 \cdot 10 \cdot 9!}{9!} = 10 \cdot 10 = \boxed{100}.$$

Pitfall

Do not compute $11!$, $10!$, and $9!$ separately. The expression is designed for cancellation.

Floor functions

The floor function gives the greatest integer less than or equal to a number.

Definition

Floor function.

The expression

$$\lfloor x \rfloor$$

means the greatest integer less than or equal to x .

Example 20

Evaluate

$$\lfloor 3.8 \rfloor, \quad \lfloor 7 \rfloor, \quad \lfloor -1.2 \rfloor.$$

Solution. The greatest integer less than or equal to 3.8 is 3, so

$$\lfloor 3.8 \rfloor = 3.$$

The greatest integer less than or equal to 7 is 7, so

$$\lfloor 7 \rfloor = 7.$$

The number -1.2 lies between -2 and -1 . The integer less than or equal to it is -2 , so

$$\lfloor -1.2 \rfloor = -2.$$

Pitfall

The floor function is not rounding toward zero. For example,

$$\lfloor -1.2 \rfloor = -2,$$

not -1 .

Floor functions on the AMC

Example 21 (2016 AMC 10A, Problem 4 Variant)

For real numbers x and $y \neq 0$, define

$$\text{rem}(x, y) = x - y \left\lfloor \frac{x}{y} \right\rfloor.$$

Evaluate

$$\text{rem}\left(\frac{8}{3}, -\frac{5}{2}\right).$$

Solution. Use the definition directly:

$$\text{rem}\left(\frac{8}{3}, -\frac{5}{2}\right) = \frac{8}{3} - \left(-\frac{5}{2}\right) \left\lfloor \frac{\frac{8}{3}}{-\frac{5}{2}} \right\rfloor.$$

Now simplify inside the floor:

$$\frac{\frac{8}{3}}{-\frac{5}{2}} = \frac{8}{3} \cdot \left(-\frac{2}{5}\right) = -\frac{16}{15}.$$

Since

$$-2 < -\frac{16}{15} < -1,$$

we get

$$\left\lfloor -\frac{16}{15} \right\rfloor = -2.$$

Substitute this back:

$$\text{rem}\left(\frac{8}{3}, -\frac{5}{2}\right) = \frac{8}{3} - \left(-\frac{5}{2}\right)(-2).$$

Now

$$\left(-\frac{5}{2}\right)(-2) = 5.$$

Therefore

$$\frac{8}{3} - 5 = \frac{8}{3} - \frac{15}{3} = \boxed{-\frac{7}{3}}.$$

Pitfall

When a problem defines a function, use the definition from the problem. Do not assume it behaves exactly like a familiar operation with the same name.

Summary

Main idea/Fact

Computational checklist.

- If there are nested fractions, work from the deepest layer outward.
- If there are grouping symbols, simplify the innermost group first.
- If there are exponents, check the base carefully.
- If there is $\sqrt{x^2}$, replace it with $|x|$, not automatically with x .
- If there is a defined operator, apply the definition every time.
- If there are factorials, cancel before multiplying.
- If there is a floor function, be careful with negative numbers.
- After each operation, rewrite the whole expression.

Problems

1. (2020 AMC 10B #1) What is the value of

$$1 - (-2) - 3 - (-4) - 5 - (-6)?$$

(A) -20 (B) -3 (C) 3 (D) 5 (E) 21

([Hint 1](#), [Hint 2](#))

2. (2018 AMC 10A #1) What is the value of

$$\left(\left((2+1)^{-1} + 1 \right)^{-1} + 1 \right)^{-1} + 1?$$

(A) $\frac{5}{8}$ (B) $\frac{11}{7}$ (C) $\frac{8}{5}$ (D) $\frac{18}{11}$ (E) $\frac{15}{8}$

([Hint 1](#), [Hint 2](#))

3. (2022 AMC 10A #1) What is the value of

$$3 + \frac{1}{3 + \frac{1}{3 + \frac{1}{3}}}$$

(A) $\frac{31}{10}$ (B) $\frac{49}{15}$ (C) $\frac{33}{10}$ (D) $\frac{109}{33}$ (E) $\frac{15}{4}$

([Hint 1](#), [Hint 2](#))

4. (2010 AMC 10B #4) For a real number x , define $\heartsuit(x)$ to be the average of x and x^2 . What is $\heartsuit(1) + \heartsuit(2) + \heartsuit(3)$?

(A) 3 (B) 6 (C) 10 (D) 12 (E) 20

([Hint 1](#), [Hint 2](#))

5. (2020 AMC 10A #3) Assuming $a \neq 3$, $b \neq 4$, and $c \neq 5$, what is the value in simplest form of the following expression?

$$\frac{a-3}{5-c} \cdot \frac{b-4}{3-a} \cdot \frac{c-5}{4-b}$$

(A) -1 (B) 1 (C) $\frac{abc}{60}$ (D) $\frac{1}{abc}$ (E) $\frac{1}{60}$

([Hint 1](#), [Hint 2](#))

6. (2016 AMC 10B #1) What is the value of $\frac{2a^{-1} + \frac{a^{-1}}{2}}{a}$ when $a = \frac{1}{2}$?

(A) 1 (B) 2 (C) $\frac{5}{2}$ (D) 10 (E) 20

([Hint 1](#), [Hint 2](#))

7. (2015 AMC 10A #1) What is the value of

$$\left(2^0 - 1 + 5^2 + 0\right)^{-1} \times 5?$$

- (A) -125 (B) -120 (C) $\frac{1}{5}$ (D) $\frac{5}{24}$ (E) 25

([Hint 1](#), [Hint 2](#))

8. (2017 AMC 10A #1) What is the value of

$$2(2(2(2(2(2+1)+1)+1)+1)+1)+1?$$

- (A) 70 (B) 97 (C) 127 (D) 159 (E) 729

([Hint 1](#), [Hint 2](#))

9. (2021 Spring AMC 10A #1) What is the value of

$$\left(2^2 - 2\right) - \left(3^2 - 3\right) + \left(4^2 - 4\right)?$$

- (A) 1 (B) 2 (C) 5 (D) 8 (E) 12

([Hint 1](#), [Hint 2](#))

10. (2021 Fall AMC 10B #3) The expression $\frac{2021}{2020} - \frac{2020}{2021}$ is equal to the fraction $\frac{p}{q}$, in which p and q are positive integers whose greatest common divisor is 1. What is p ?

- (A) 1 (B) 9 (C) 2020 (D) 2021 (E) 4041

([Hint 1](#), [Hint 2](#))

Hints

Hint 1

1. **MAIN IDEA:** Subtracting a negative is the same as adding a positive. Rewrite the entire expression by converting every $-(-k)$ into $+k$ before you compute anything.

2. **MAIN IDEA:** Recall that $x^{-1} = \frac{1}{x}$. Start from the innermost parentheses and work outward, converting each $(\dots)^{-1}$ into a fraction $\frac{1}{\dots}$ as you go.

3. **MAIN IDEA:** A continued fraction is evaluated from the bottom up. Start with the innermost fraction $\frac{1}{3}$ and work your way out one layer at a time.

4. **MAIN IDEA:** A custom operator is just a recipe—substitute and simplify. The average of two numbers p and q is $\frac{p+q}{2}$. Write out $\heartsuit(x) = \frac{x+x^2}{2}$ explicitly, then plug in $x = 1, 2, 3$ one at a time.

5. **MAIN IDEA:** Before multiplying anything out, look at the numerators and denominators across all three fractions. Can you pair each numerator with a denominator from a different fraction?

6. **MAIN IDEA:** Rewrite all negative exponents as fractions first. Since $a = \frac{1}{2}$, we have $a^{-1} = \frac{1}{a} = 2$. Substitute $a^{-1} = 2$ into the numerator before doing anything else.

7. **MAIN IDEA:** Do the inside of the parentheses first, but do not forget what the exponent -1 means. What is $2^0 - 1 + 5^2 + 0$ before applying the power -1 ?

8. **MAIN IDEA:** Nested expressions are evaluated from the inside out. Start with $2 + 1$, then each new layer has the form $2(\text{previous value}) + 1$.

9. **MAIN IDEA:** Evaluate each parenthesized piece separately first. What are $2^2 - 2$, $3^2 - 3$, and $4^2 - 4$?

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10. **MAIN IDEA:** Combine the two fractions using a common denominator. What is the numerator when both fractions are written over $2020 \cdot 2021$?

Hint 2

1. After rewriting, you should have $1 + 2 - 3 + 4 - 5 + 6$. Now add and subtract from left to right. Group the terms strategically: $(1 + 2) + (-3 + 4) + (-5 + 6)$.

2. After the first step you should have $\left(\left(\frac{1}{3} + 1\right)^{-1} + 1\right)^{-1} + 1$. Simplify $\frac{1}{3} + 1 = \frac{4}{3}$. Then take its inverse: $\left(\frac{4}{3}\right)^{-1} = \frac{3}{4}$. Keep going one layer at a time.

3. The innermost fraction gives $3 + \frac{1}{3} = \frac{10}{3}$. Now the next layer is $3 + \frac{1}{\frac{10}{3}}$. To divide by a fraction, multiply by its reciprocal: $\frac{1}{\frac{10}{3}} = \frac{3}{10}$.

4. You should get $\heartsuit(1) = \frac{1+1}{2} = 1$, $\heartsuit(2) = \frac{2+4}{2} = 3$, and $\heartsuit(3) = \frac{3+9}{2} = 6$. Now just add them up.

5. Notice that $3 - a = -(a - 3)$ and $5 - c = -(c - 5)$ and $4 - b = -(b - 4)$. Each numerator appears in a denominator but with opposite sign. Count how many negative signs you pick up from flipping.

6. After substituting $a^{-1} = 2$, the numerator becomes $2(2) + \frac{2}{2} = 4 + 1 = 5$. Now you need to divide by $a = \frac{1}{2}$. Dividing by $\frac{1}{2}$ is the same as multiplying by 2.

7. The inside becomes

$$2^0 - 1 + 5^2 + 0 = 1 - 1 + 25 + 0 = 25.$$

So the expression becomes

$$25^{-1} \times 5.$$

Now use $25^{-1} = \frac{1}{25}$.

8. Starting from the inside, the values go

$$2 + 1 = 3, \quad 2(3) + 1 = 7, \quad 2(7) + 1 = 15.$$

Keep applying the same rule until you reach the outside.

9. After simplifying the three parenthesized pieces, the expression becomes

$$(4 - 2) - (9 - 3) + (16 - 4) = 2 - 6 + 12.$$

Now add and subtract from left to right.

10. Combining gives

$$\frac{2021}{2020} - \frac{2020}{2021} = \frac{2021^2 - 2020^2}{2020 \cdot 2021}.$$

Now use

$$2021^2 - 2020^2 = (2021 - 2020)(2021 + 2020).$$

Word Problems and Conversions

Introduction

So far every expression told us exactly what to do. The AMC does not always hand you an expression. Sometimes it hands you a story:

Example 22 (2021 Spring AMC 10B, Problem 4)

At a math contest, 57 students are wearing blue shirts, and another 75 students are wearing yellow shirts. The 132 students are assigned into 66 pairs. In exactly 23 of these pairs, both students are wearing blue shirts. In how many pairs are both students wearing yellow shirts?

(A) 23 (B) 32 (C) 37 (D) 41 (E) 64

Solution. There are three kinds of pairs: blue-blue, blue-yellow, and yellow-yellow. We know there are 66 pairs in total and we know 23 of them are blue-blue. So if we can count the blue-yellow pairs, the yellow-yellow pairs are whatever is left.

Every blue-blue pair uses up two blue students, so those 23 pairs use

$$23 \cdot 2 = 46$$

blue students. That leaves

$$57 - 46 = 11$$

blue students, and each of them must be paired with a yellow student. So there are exactly 11 blue-yellow pairs.

Now subtract:

$$66 - 23 - 11 = 32.$$

Therefore the answer is

(B) 32.

Remark. Notice what we never did. We never introduced a variable and we never wrote an equation. We turned each sentence into a number and kept track of what was left over. A surprising number of early AMC problems fall to exactly this.

That will not always be enough. Sometimes the story hides a quantity we cannot compute directly, and then we need a name for it. This lecture covers both situations.

The main habit

A **word problem** is not a different kind of mathematics. It is the same arithmetic and algebra you already know, written in English instead of symbols. The whole difficulty is the translation.

Main idea/Fact (Main habit for word problems)

Read one sentence. Write one line of mathematics. Then read the next sentence.

Do not try to hold the whole story in your head and solve it at the end.

Students lose these problems in two places: they read the whole paragraph at once and panic, or they compute something true but not the thing that was asked. Both are fixed by the same discipline of going sentence by sentence and writing as you go.

Start from the question

Before translating anything, read the last sentence. That tells you what you are hunting for, and it usually tells you what you need in order to get it.

Main idea/Fact (Work backwards from the question)

Ask what the final sentence wants. Then ask what you would need in order to produce it. Then ask what you would need in order to produce that.

Keep going until you reach something the problem gives you directly.

Example 23

You have 3 brothers and 1 sister. Your sister has \$3, each of your brothers has \$5, and you have \$2. Together you want to buy a cake that costs \$30. Can you afford it?

Solution. The question asks whether the group can afford a \$30 cake. To answer that, we need the total amount of money the group has. To get the total, we need each person's amount, and all of those are given.

Your brothers have

$$3 \cdot 5 = 15$$

dollars between them. Adding your sister and yourself:

$$15 + 3 + 2 = 20.$$

Since

$$20 < 30,$$

the group cannot afford the cake.

The chain we followed was: can we afford it, so we need the total, so we need each person. That is the backward reading, and every step of it was forced by the final sentence.

Naming the unknown

Sometimes the quantity you need is not given and cannot be computed in one step. Give it a name and carry on as if you knew it. A named **unknown** behaves exactly like a number: you can add it, multiply it, and compare it.

Main idea/Fact (Labelling)

If a quantity is unknown but you need to talk about it, call it x , or better, a letter that reminds you what it is.

Example 24

In the situation above, how much more money does the group need?

Solution. Call the missing amount x . The group currently has 20 dollars, and we want the total to reach 30:

$$20 + x = 30.$$

Therefore

$$x = \boxed{10}.$$

Choosing good letters is worth a moment of thought. Writing b for blueberry and c for cherry costs nothing and saves you from solving for the wrong variable three lines later.

The translation dictionary

Certain English phrases mean exactly one thing in mathematics. Learn them so the translation is automatic.

Main idea/Fact (Useful phrases) • “more than” means add

- “less than” means subtract
- “is”, “was”, “has” means $=$
- “times”, “as many as” means multiply
- “twice” means multiply by 2
- “30% of x ” means $0.3x$
- “half of x ”, “two thirds of x ” mean $\frac{1}{2}x$, $\frac{2}{3}x$
- “the rest” means the total minus everything already accounted for

Example 25

Translate each sentence into mathematics.

- (a) Alice has 3 more books than Bob.
- (b) The number of mangoes is 4 times the number of apples.
- (c) A number is 3 less than 4 times another number.
- (d) Father gave money to me, my brother, and my sister. My brother took 25% of it, I took 70% of my brother’s amount, and my sister took the rest.
- (e) Take a number. Take half of it. Then take two thirds of the new number. Then take one fourth of the new number. Finally take five twelfths of the new number.

Solution.

- (a) Let a be Alice’s books and b be Bob’s. The word “has” is the equals sign, and “more than” is addition:

$$a = b + 3.$$

- (b) Let m be mangoes and a be apples:

$$m = 4a.$$

- (c) Let the two numbers be x and y :

$$x = 4y - 3.$$

(d) Let the total be x . My brother took

$$0.25x.$$

I took 70% of *his* amount, not of the total:

$$0.7 \cdot 0.25x = 0.175x.$$

My sister took the rest, which is the total minus both:

$$x - 0.25x - 0.175x = 0.575x.$$

(e) Let the number be x . Half of it is

$$\frac{1}{2}x.$$

Two thirds of that is

$$\frac{2}{3} \cdot \frac{1}{2}x = \frac{1}{3}x.$$

One fourth of that is

$$\frac{1}{4} \cdot \frac{1}{3}x = \frac{1}{12}x.$$

Five twelfths of that is

$$\frac{5}{12} \cdot \frac{1}{12}x = \frac{5}{144}x.$$

Pitfall

English reverses the order of subtraction. “3 less than $4y$ ” is

$$4y - 3,$$

not

$$3 - 4y.$$

The same trap appears with “less than” in comparisons, so read those phrases twice.

Pitfall

In part (d), “70% of my brother’s amount” is a percent of a percent. Taking 70% of the total instead of 70% of $0.25x$ is the most common error on this kind of sentence. Always ask: a percent *of what*?

Exercise 26. Translate and solve: twice a number, decreased by 5, equals the number increased by 4.¹

Exercise 27. A shirt costs \$40 after a 20% discount. What was the original price?²

¹ $2x - 5 = x + 4$, so $x = 9$.

²If p is the original price then $0.8p = 40$, so $p = 50$. It is not $40 \cdot 1.2 = 48$, because the discount was taken off the original price, not off \$40.

Practising the translation

The translation is a skill on its own, separate from solving. Here is one worth doing slowly.

Example 28

When a number is divided by 4 and then 3 is added, the result is the same as 7 less than the sum of the number and 1. Find the number.

Solution. Let the number be x . The left side of the sentence says divide by 4, then add 3:

$$\frac{x}{4} + 3.$$

The right side says take the sum of the number and 1, then subtract 7:

$$(x + 1) - 7.$$

The phrase “is the same as” is the equals sign:

$$\frac{x}{4} + 3 = (x + 1) - 7.$$

Simplify the right side:

$$\frac{x}{4} + 3 = x - 6.$$

Subtract $\frac{x}{4}$ from both sides and add 6 to both sides:

$$9 = \frac{3x}{4}.$$

Therefore

$$x = \boxed{12}.$$

Fix your units first

A word problem will happily mix dollars with cents, minutes with seconds, and miles with feet. An equation cannot.

Main idea/Fact (One unit per equation)

Before writing any equation, choose one unit for each kind of quantity and convert everything into it.

Money with coins goes to cents. Time with short intervals goes to seconds. Length with a mile in it goes to feet.

Example 29

A jar contains n quarters and m dimes, worth \$4.05 in total. Write the equation.

Solution. A quarter is 25 cents and a dime is 10 cents, so the natural unit here is cents. Converting the total,

$$\$4.05 = 405 \text{ cents},$$

and the equation is

$$25n + 10m = 405.$$

Pitfall

Writing

$$25n + 10m = 4.05$$

mixes cents on the left with dollars on the right, and it is wrong by a factor of 100. Convert before you write, not after you solve.

The same quantity, described twice

Here is the single most useful move for building an equation. If the problem describes one quantity in two different ways, those two descriptions are equal, and that equality is your equation.

Main idea/Fact (Count it twice)

Look for a quantity the problem mentions from two directions: the same money, the same distance, the same water, the same number of objects.

Write it once each way and set the two expressions equal.

Example 30

Tray A holds 3 rows of 8 cookies. Tray B holds 4 equal rows and contains exactly as many cookies as tray A. How many cookies are in each row of tray B?

Solution. Let r be the number of cookies in each row of tray B. The number of cookies on tray A is

$$3 \cdot 8 = 24,$$

and the number on tray B is

$$4r.$$

These count the same quantity, so

$$4r = 24,$$

and therefore

$$r = \boxed{6}.$$

This looks trivial here because both descriptions are easy. On the AMC one of the two descriptions is usually buried in a later sentence, and finding it is the whole problem.

Digits and value

Contest problems often talk about the digits of a number. A digit and a number are not the same object, and mixing them up ruins the equation.

Definition (Place value)

If a two digit number has tens digit a and units digit b , the number itself is

$$10a + b.$$

If a three digit number has digits a, b, c from left to right, the number is

$$100a + 10b + c.$$

Example 31

A two digit number has digit sum 11, and reversing its digits increases it by 27. Find the number.

Solution. Let the digits be a and b , so the number is $10a + b$. The digit sum is a statement about digits:

$$a + b = 11.$$

The reversed number is $10b + a$, and it is 27 larger:

$$10b + a = (10a + b) + 27.$$

Simplify the second equation:

$$9b - 9a = 27,$$

so

$$b - a = 3.$$

Adding this to $a + b = 11$ gives $2b = 14$, so $b = 7$ and $a = 4$. The number is

$$\boxed{47}.$$

Checking against the words: $4 + 7 = 11$, and $74 - 47 = 27$.

Pitfall

Writing the two digit number as ab means a times b , which is not what you want. Always expand into $10a + b$ before doing any algebra.

One unknown, one equation

Now we combine naming and translating on a real contest problem.

Example 32 (2017 AMC 10B, Problem 5)

Camilla had twice as many blueberry jelly beans as cherry jelly beans. After eating 10 pieces of each kind, she now has three times as many blueberry jelly beans as cherry jelly beans. How many blueberry jelly beans did she originally have?

- (A) 10 (B) 20 (C) 30 (D) 40 (E) 50

Solution. Let b be the original blueberry count and c the original cherry count. The first sentence says

$$b = 2c.$$

The second sentence describes the counts *after* eating 10 of each, which are $b - 10$ and $c - 10$:

$$b - 10 = 3(c - 10).$$

Substitute $b = 2c$ into the second equation:

$$2c - 10 = 3c - 30.$$

So

$$c = 20,$$

and therefore

$$b = 2 \cdot 20 = \boxed{\text{(D)} 40}.$$

Main idea/Fact (Before and after)

When a problem changes the situation partway through, write the “after” quantities in terms of the “before” quantities. Do not give them brand new letters.

Two unknowns, two equations

If the story contains two unknowns, expect two facts. Each fact becomes an equation.

Main idea/Fact (Constructing a system)

- Read each sentence separately.
- Name every unknown quantity.
- Ask how that sentence relates the unknowns to each other.

Count as you go. Two unknowns need two independent equations.

Example 33 (2021 AMC 10A, Problem 2)

Portia’s high school has 3 times as many students as Lara’s high school. The two high schools have a total of 2600 students. How many students does Portia’s high school have?

(A) 600 (B) 650 (C) 1950 (D) 2000 (E) 2050

Solution. Let p be the number of students at Portia’s school and l the number at Lara’s. The first sentence is a “times” sentence:

$$p = 3l.$$

The second sentence is a total:

$$p + l = 2600.$$

From the first equation, $l = \frac{p}{3}$. Substituting into the second:

$$p + \frac{p}{3} = 2600.$$

So

$$\frac{4p}{3} = 2600,$$

and therefore

$$p = \boxed{\text{(C)} 1950}.$$

Pitfall

“Portia’s school has 3 times as many students as Lara’s” is

$$p = 3l,$$

not

$$l = 3p.$$

Sanity check the direction: Portia’s school should be the bigger one. If your equation makes the wrong school bigger, you flipped it.

Percent inside a system

Percent sentences translate the same way, they just produce decimals.

Example 34 (2021 AMC 10B, Problem 3)

In an after-school program for juniors and seniors, there is a debate team with an equal number of students from each class on the team. Among the 28 students in the program, 25% of the juniors and 10% of the seniors are on the debate team. How many juniors are in the program?

(A) 5 (B) 6 (C) 8 (D) 11 (E) 20

Solution. Let j be the number of juniors and s the number of seniors in the program. The program has 28 students:

$$j + s = 28.$$

The number of juniors on the debate team is 25% of j , and the number of seniors on it is 10% of s . The team has equally many of each:

$$0.25j = 0.1s.$$

Multiply both sides by 20 to clear decimals:

$$5j = 2s.$$

From the first equation $s = 28 - j$, so

$$5j = 2(28 - j).$$

Expanding gives

$$5j = 56 - 2j,$$

so $7j = 56$ and

$$j = \boxed{\text{(C) } 8}.$$

Pitfall

The percents here are taken of *different* totals: 25% of the juniors and 10% of the seniors. You may not add or compare the percents directly. Convert each into an actual count first.

Ages and time shifts

Age problems are pure translation, and they have one specific trap.

Example 35

A father is three times as old as his son. In 8 years he will be twice as old as his son. How old is each now?

Solution. Let f be the father's current age and s the son's current age. The first sentence gives

$$f = 3s.$$

In 8 years, *both* people are 8 years older, so their ages are $f + 8$ and $s + 8$:

$$f + 8 = 2(s + 8).$$

Substitute $f = 3s$:

$$3s + 8 = 2s + 16.$$

So

$$s = 8,$$

and therefore

$$f = 24.$$

Pitfall

Time passes for everyone. Writing

$$f + 8 = 2s$$

forgets to age the son, and it is the single most common mistake on these problems.

Rates

A **rate** problem gives you a repeating process. Find the amount per single unit of time first, then scale.

Main idea/Fact (Find the per-unit amount)

Compute how much happens in one round, one second, or one day. Everything else follows by multiplying or dividing.

Example 36

A machine fills 3 bottles every 5 seconds. How long does it take to fill 42 bottles?

Solution. In one second the machine fills

$$\frac{3}{5}$$

of a bottle. To fill 42 bottles we need

$$\frac{42}{3/5} = 42 \cdot \frac{5}{3} = 70$$

seconds.

Equivalently, 42 bottles is 14 groups of 3, and each group takes 5 seconds, giving $14 \cdot 5 = 70$.

When the last step is different

The scaling above assumed every round is identical. Read carefully, because sometimes the final round is not.

Example 37 (2017 AMC 10A, Problem 4)

Mia is “helping” her mom pick up 30 toys that are strewn on the floor. Mia’s mom manages to put 3 toys into the toy box every 30 seconds, but each time immediately after those 30 seconds have elapsed, Mia takes 2 toys out of the box. How much time, in minutes, will it take Mia and her mom to put all 30 toys into the box for the first time?

(A) 13.5 (B) 14 (C) 14.5 (D) 15 (E) 15.5

Solution. In each 30 second round, mom adds 3 toys and Mia removes 2, for a net gain of

$$3 - 2 = 1$$

toy per round.

Now be careful about the ending. The moment the box first contains all 30 toys, the job is finished and Mia never gets to remove anything. So the final round is not a net gain of 1, it is a gain of 3.

That means the net rounds only need to bring the box to 27 toys, which takes 27 rounds. Then one more round puts in the last 3 toys. The total time is

$$27 \cdot 30 + 30 = 840$$

seconds, which is

$$\frac{840}{60} = \boxed{\text{(B)} 14}$$

minutes.

Pitfall

Dividing 30 by the net rate of 1 toy per round gives 30 rounds, or 15 minutes, which is answer (D). That answer is on the list precisely because it is the natural mistake.

The same care is needed when counting objects rather than rounds.

Main idea/Fact (Closure)

If n objects are placed in a row, there are only

$$n - 1$$

gaps between them.

Counting n gaps instead of $n - 1$ is called a **fence-post error**.

Exercise 38. Twelve fence posts stand in a straight line, equally spaced, and the distance from the first post to the last is 66 feet. How far apart are consecutive posts?³

Using and checking the answers

Every AMC answer is one of five given numbers. That has two consequences.

Main idea/Fact (Test the choices)

If setting up the equation is hard but checking a value is easy, work backwards from the answer choices instead.

This is especially strong when the answer is a small integer, or when the question asks for the greatest or smallest value with some property.

Example 39

Confirm the jelly bean answer by testing choices instead of solving.

Solution. Test (D) 40. Then she had 40 blueberry and, since that is twice the cherry count, 20 cherry. After eating 10 of each she has 30 and 10, and

$$30 = 3 \cdot 10,$$

so (D) works.

Test (B) 20 for contrast. Then she had 20 blueberry and 10 cherry. After eating 10 of each she has 10 and 0, and 10 is not three times 0. So (B) fails.

The second consequence is that the wrong answers were designed by someone who knows exactly which mistake you are about to make. Finding your value on the list is not confirmation.

Main idea/Fact (Final check)

Substitute your answer back into the original sentences and confirm each one.

Then reread the question itself. Problems frequently ask for a quantity you did not solve for: the total instead of one part, the difference instead of a value, or a ratio instead of either number.

³Twelve posts leave 11 gaps, so each gap is $66/11 = 6$ feet. It is not $66/12$.

Summary

Main idea/Fact (Setting up) • Read the last sentence first and identify exactly what is asked.

- If nothing except the question is unknown, compute forward and use no variables at all.
- Otherwise name each unknown with a letter that reminds you what it is.
- Translate one sentence into one line of mathematics, then move to the next sentence.
- Convert everything to a single unit before writing any equation.
- Count as you go: k unknowns need k independent equations. Hunt for a quantity described twice.

Main idea/Fact (Finishing) • Write “after” quantities in terms of “before” quantities.

- Ask of every percent: a percent *of what?*
- Expand digit problems into $10a + b$ before doing algebra.
- Watch closure: n objects in a row leave $n - 1$ gaps, and a final step may behave differently from the repeating ones.
- Reread the question, then substitute your answer back into the original sentences.

Problems

These are not sorted by difficulty. Try as many as you can before looking at the hints.

1. (2025 AMC 10A #2/12A #2) A box contains 10 pounds of a nut mix that is 50 percent peanuts, 20 percent cashews, and 30 percent almonds. A second nut mix containing 20 percent peanuts, 40 percent cashews, and 40 percent almonds is added to the box resulting in a new nut mix that is 40 percent peanuts. How many pounds of cashews are now in the box?

(A) 3.5 (B) 4 (C) 4.5 (D) 5 (E) 6

(Hint 1, Hint 2)

2. (2023 AMC 10B #1/12B #1) Mrs. Jones is pouring orange juice for her 4 kids into 4 identical glasses. She fills the first 3 full, but only has enough orange juice to fill one third of the last glass. What fraction of a glass of orange juice does she need to pour from the 3 full glasses into the last glass so that all glasses have an equal amount of orange juice?

(A) $\frac{1}{12}$ (B) $\frac{1}{4}$ (C) $\frac{1}{6}$ (D) $\frac{1}{8}$ (E) $\frac{2}{9}$

(Hint 1, Hint 2)

3. (2023 AMC 10B #2) Carlos went to a sports store to buy running shoes. Running shoes were on sale, with prices reduced by 20% on every pair of shoes. Carlos also knew that he had to pay a 7.5% sales tax on the discounted price. He had 43 dollars. What is the original (before discount) price of the most expensive shoes he could afford to buy?

(A) 46 (B) 50 (C) 48 (D) 47 (E) 49

(Hint 1, Hint 2)

4. (2021 Fall AMC 10A #6/12A #5) Elmer the emu takes 44 equal strides to walk between consecutive telephone poles on a rural road. Oscar the ostrich can cover the same distance in 12 equal leaps. The telephone poles are evenly spaced, and the 41st pole along this road is exactly one mile (5280 feet) from the first pole. How much longer, in feet, is Oscar's leap than Elmer's stride?

(A) 6 (B) 8 (C) 10 (D) 11 (E) 15

(Hint 1, Hint 2)

5. (2019 AMC 10B #1/12B #1) Alicia had two containers. The first was $\frac{5}{6}$ full of water and the second was empty. She poured all the water from the first container into the second container, at which point the second container was $\frac{3}{4}$ full of water. What is the ratio of the volume of the first container to the volume of the second container?

(A) $\frac{5}{8}$ (B) $\frac{4}{5}$ (C) $\frac{7}{8}$ (D) $\frac{9}{10}$ (E) $\frac{11}{12}$

(Hint 1, Hint 2)

6. (2019 AMC 10B #3) In a high school with 500 students, 40% of the seniors play a musical instrument, while 30% of the non-seniors do not play a musical instrument. In all, 46.8% of the students do not play a musical instrument. How many non-seniors play a musical instrument?

(A) 66 (B) 154 (C) 186 (D) 220 (E) 266

[\(Hint 1, Hint 2\)](#)

7. (2019 AMC 10B #7/12B #5) Each piece of candy in a store costs a whole number of cents. Casper has exactly enough money to buy either 12 pieces of red candy, 14 pieces of green candy, 15 pieces of blue candy, or n pieces of purple candy. A piece of purple candy costs 20 cents. What is the smallest possible value of n ?

(A) 18 (B) 21 (C) 24 (D) 25 (E) 28

[\(Hint 1, Hint 2\)](#)

8. (2018 AMC 10A #6) Sangho uploaded a video to a website where viewers can vote that they like or dislike a video. Each video begins with a score of 0, and the score increases by 1 for each like vote and decreases by 1 for each dislike vote. At one point Sangho saw that his video had a score of 90, and that 65% of the votes cast on his video were like votes. How many votes had been cast on Sangho's video at that point?

(A) 200 (B) 300 (C) 400 (D) 500 (E) 600

[\(Hint 1, Hint 2\)](#)

9. (2018 AMC 10A #8) Joe has a collection of 23 coins, consisting of 5-cent coins, 10-cent coins, and 25-cent coins. He has 3 more 10-cent coins than 5-cent coins, and the total value of his collection is 320 cents. How many more 25-cent coins does Joe have than 5-cent coins?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

[\(Hint 1, Hint 2\)](#)

10. (2017 AMC 10A #2/12A #1) Pablo buys popsicles for his friends. The store sells single popsicles for \$1 each, 3-popsicle boxes for \$2, and 5-popsicle boxes for \$3. What is the greatest number of popsicles that Pablo can buy with \$8?

(A) 8 (B) 11 (C) 12 (D) 13 (E) 15

[\(Hint 1, Hint 2\)](#)

11. (2016 AMC 10A #3) For every dollar Ben spent on bagels, David spent 25 cents less. Ben paid \$12.50 more than David. How much did they spend in the bagel store together?

(A) \$37.50 (B) \$50.00 (C) \$87.50 (D) \$90.00 (E) \$92.50

[\(Hint 1, Hint 2\)](#)

12. (2016 AMC 10A #8) Trickster Rabbit agrees with Foolish Fox to double Fox's money every time Fox crosses the bridge by Rabbit's house, as long as Fox pays 40 coins in toll to Rabbit after each crossing. The payment is made after the doubling, Fox is excited about his good fortune until he discovers that all his money is gone after crossing the bridge three times. How many coins did Fox have at the beginning?

(A) 20 (B) 30 (C) 35 (D) 40 (E) 45

[\(Hint 1, Hint 2\)](#)

13. (2015 AMC 10A #4) Pablo, Sofia, and Mia got some candy eggs at a party. Pablo had three times as many eggs as Sofia, and Sofia had twice as many eggs as Mia. Pablo decides to give some of his eggs to Sofia and Mia so that all three will have the same number of eggs. What fraction of his eggs should Pablo give to Sofia?

- (A) $\frac{1}{12}$ (B) $\frac{1}{6}$ (C) $\frac{1}{4}$ (D) $\frac{1}{3}$ (E) $\frac{1}{2}$

([Hint 1](#), [Hint 2](#))

14. (2014 AMC 10A #2) Roy's cat eats $\frac{1}{3}$ of a can of cat food every morning and $\frac{1}{4}$ of a can of cat food every evening. Before feeding his cat on Monday morning, Roy opened a box containing 6 cans of cat food. On what day of the week did the cat finish eating all the cat food in the box?

- (A) Tuesday (B) Wednesday (C) Thursday (D) Friday (E) Saturday

([Hint 1](#), [Hint 2](#))

15. (2014 AMC 10A #3) Bridget bakes 48 loaves of bread for her bakery. She sells half of them in the morning for \$2.50 each. In the afternoon she sells two thirds of what she has left, and because they are not fresh, she charges only half price. In the late afternoon she sells the remaining loaves at a dollar each. Each loaf costs \$0.75 for her to make. In dollars, what is her profit for the day?

- (A) 24 (B) 36 (C) 44 (D) 48 (E) 52

([Hint 1](#), [Hint 2](#))

16. (2012 AMC 10B #5) Anna enjoys dinner at a restaurant in Washington, D.C., where the sales tax on meals is 10%. She leaves a 15% tip on the prices of her meal before the sales tax is added, and the tax is calculated on the pre-tip amount. She spends a total of 27.50 for dinner. What is the cost of her dinner without tax or tip?

- (A) \$18 (B) \$20 (C) \$21 (D) \$22 (E) \$24

([Hint 1](#), [Hint 2](#))

17. (2012 AMC 12B #3) For a science project, Sammy observed a chipmunk and a squirrel stashing acorns in holes. The chipmunk hid 3 acorns in each of the holes it dug. The squirrel hid 4 acorns in each of the holes it dug. They each hid the same number of acorns, although the squirrel needed 4 fewer holes. How many acorns did the chipmunk hide?

- (A) 30 (B) 36 (C) 42 (D) 48 (E) 54

([Hint 1](#), [Hint 2](#))

Hints

Hint 1

1. **MAIN IDEA:** The only unknown is how many pounds of the second mix are added, so call that x . Before you use x at all, turn every percent in the first box into an actual weight in pounds. How many pounds of peanuts, cashews, and almonds start in the box?

2. **MAIN IDEA:** Pouring juice between glasses does not create or destroy juice, so the total is fixed. In glasses' worth, how much juice does she have altogether, and therefore how much ends up in each of the 4 glasses?

3. **MAIN IDEA:** Translate the sentence order exactly: the discount happens first, then the tax is charged on the discounted price. If x is the original price, what is the discounted price in terms of x ?

4. **MAIN IDEA:** Both animals cover the same distance, namely the gap between two consecutive poles, so find that gap first. Between the 1st pole and the 41st pole, how many gaps are there? It is not 41.

5. **MAIN IDEA:** The same water is described twice, and that repetition is your equation. Let f and s be the two volumes, then write the amount of water once as a part of f and once as a part of s .

6. **MAIN IDEA:** Let s be the number of seniors and n the number of non-seniors, so $s + n = 500$. The 46.8% fact is about students who do *not* play. Can you restate the senior fact so that it is also about not playing?

7. **MAIN IDEA:** “Exactly enough money” is a divisibility statement. If M is Casper’s money in cents, what does being able to buy exactly 12 red candies tell you about M ?

8. **MAIN IDEA:** Let x be the total number of votes. The score is likes minus dislikes, not the number of likes. If 65% of the votes are likes, what percent are dislikes?

9. **MAIN IDEA:** Three unknowns need three equations, and the problem gives you exactly three facts: the number of coins, the comparison between two of the counts, and the total value. Let x , y , z be the numbers of 5, 10, and 25 cent coins and write all three. Work in cents throughout.

10. **MAIN IDEA:** This is a value comparison, so measure each option by popsicles per dollar. Which of the three options is the best deal by that measure?

11. **MAIN IDEA:** The phrase “for every dollar Ben spent” sets up a fixed ratio. For each dollar Ben spends, David spends 75 cents, so the gap between them grows by 25 cents. How many of those 25 cent gaps add up to \$12.50?

12. **MAIN IDEA:** Forward is messy because the starting amount is unknown, but backward is clean because the ending amount is 0. Undo the last crossing: what did Fox have just before paying the 40 coin toll, and therefore just before the doubling?

13. **MAIN IDEA:** Put everyone in terms of the smallest quantity. Let Mia have x eggs, then write Sofia's and Pablo's counts in terms of x . How many eggs should each person end up with?

14. **MAIN IDEA:** Find the rate first. How much of a can does the cat eat over one full day, morning plus evening?

15. **MAIN IDEA:** Nothing here is unknown, so compute forward one sentence at a time and never introduce a variable. After the morning sale, how many loaves are left and how much money has come in?

16. **MAIN IDEA:** The question asks for the pre-tax, pre-tip meal price, so name it x . Both the tip and the tax are computed from that same x . What are they in terms of x ?

17. **MAIN IDEA:** The unknowns are the numbers of *holes*, not the numbers of acorns. Let c and s be the numbers of holes dug by the chipmunk and the squirrel. What is each animal's acorn total in terms of c and s ?

Hint 2

1. The first box holds 5 pounds of peanuts, 2 of cashews, and 3 of almonds. After adding x pounds of the second mix, the total weight is $10 + x$ and the peanut weight is $5 + 0.2x$. Set up “peanuts are 40% of the total” as one equation and solve for x . Then remember the question is about cashews, not about x .

2. The total is $3 + \frac{1}{3} = \frac{10}{3}$ glasses, so each of the 4 glasses must end with $\frac{10}{12}$ of a glass. Now compare a currently full glass to that final amount.

3. The discounted price is $0.8x$ and the tax is 7.5% of that, so Carlos pays $0.8x + 0.075(0.8x)$. The most expensive shoes he can afford make this exactly 43. Solve for x .

4. There are 40 gaps, so one gap is $\frac{5280}{40}$ feet. A stride is one gap split into 44 equal parts and a leap is one gap split into 12. Now subtract.

5. The water is $\frac{5}{6}f$ and it is also $\frac{3}{4}s$, so

$$\frac{5}{6}f = \frac{3}{4}s.$$

The question asks for $\frac{f}{s}$, so rearrange to isolate that ratio rather than solving for f and s separately.

6. If 40% of seniors play then 60% do not, and 30% of non-seniors do not. Also 46.8% of 500 is the actual number of students who do not play. That gives a second equation alongside $s + n = 500$. After solving, make sure you report non-seniors who *do* play.

7. M is a multiple of 12, of 14, and of 15, and also $M = 20n$. Making n as small as possible means making M as small as possible. What is the smallest positive number divisible by 12, 14, and 15?

8. Likes are $0.65x$ and dislikes are $0.35x$, so the score is $0.65x - 0.35x$. Set that equal to 90 and solve.

9. Your three equations should be

$$x + y + z = 23, \quad y = x + 3, \quad 5x + 10y + 25z = 320.$$

Use the first two to write z in terms of x , then substitute into the third. The question asks for $z - x$, not for z .

10. The 5-box gives $\frac{5}{3}$ popsicles per dollar, the 3-box gives $\frac{3}{2}$, and singles give 1. Buy as many of the best deal as the money allows, then spend whatever is left on the next best. Check that no money is wasted.

11. Since $\frac{12.50}{0.25} = 50$, Ben spent \$50. Now use the ratio to get David's total, and remember the question asks what they spent *together*.

12. Just before the toll he had 40 coins, so just before the doubling he had 20. That 20 is exactly what he had after the second crossing. Apply the same undo two more times.

13. Mia has x , Sofia has $2x$, Pablo has $6x$, so the total is $9x$ and everyone should end with $3x$. Work out how many eggs Sofia needs to receive, then express that as a fraction of Pablo's *original* amount.

14. The cat eats $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$ of a can per day, and 6 cans is $\frac{72}{12}$. Count off whole days until what remains is less than a full morning portion. The box runs out during a meal, not at the end of a day, so name that day carefully.

15. She sells 24 loaves at \$2.50, then two thirds of the remaining 24 at half price, then the last few at \$1. Add up the revenue, then subtract the cost of baking all 48 loaves, not just the ones sold in the morning.

16. The tip is $0.15x$ and the tax is $0.10x$, and her total is the meal plus both of those. That is a single equation in x .

17. They hid the same number of acorns, which gives $3c = 4s$. The squirrel needed 4 fewer holes, which gives $s = c - 4$. Substitute and solve, then answer the question actually asked: the number of acorns the chipmunk hid, not the number of holes.

Speed, Distance, Time and Work

Introduction

Everything in this part is a word problem, so the habits from Part 2 still apply: read the last sentence first, name the unknowns, translate one sentence at a time. What is new is that these problems are all built on a single relationship, and that relationship has a trap in it.

Example 40 (2013 AMC 10B, Problem 8)

Ray's car averages 40 miles per gallon of gasoline, and Tom's car averages 10 miles per gallon of gasoline. Ray and Tom each drive the same number of miles. What is the cars' combined rate of miles per gallon of gasoline?

(A) 10 (B) 16 (C) 25 (D) 30 (E) 40

Solution. The tempting move is to average 40 and 10 to get 25. That answer is on the list, and it is wrong.

Here is why. A combined rate is total miles divided by total gallons, so we need both totals. Let each of them drive x miles. Ray's car uses

$$\frac{x}{40}$$

gallons, and Tom's car uses

$$\frac{x}{10}$$

gallons. Together they drive $2x$ miles on

$$\frac{x}{40} + \frac{x}{10} = \frac{x}{40} + \frac{4x}{40} = \frac{5x}{40} = \frac{x}{8}$$

gallons. So the combined rate is

$$\frac{2x}{x/8} = 2x \cdot \frac{8}{x} = \boxed{\text{(B) } 16}$$

miles per gallon.

Remark. Notice that 16 is much closer to 10 than to 40. That is not an accident: Tom's car burns far more gasoline over the same distance, so his terrible mileage dominates the average. Rates behave this way constantly, and it is worth developing the instinct now.

Notice also that gallons never behaved like a quantity of fuel in that solution. They behaved exactly like time does in a speed problem. That is the real subject of this part.

The main habit

A **rate** is one quantity measured per unit of another. Speed is distance per unit of time. Fuel efficiency is distance per unit of fuel. A work rate is jobs per unit of time. All three obey the same single relationship.

Main idea/Fact (Main habit for rate problems)

Every rate problem is

$$\text{amount} = \text{rate} \times \text{time}.$$

Write down which two of the three you know, and which one the question wants.

When you cannot answer directly, go back to the two totals and divide.

That last line is what saved us above. Whenever a rate question gets confusing, the reliable move is to stop manipulating rates and go back to counting the raw amounts.

The rate triangle

Definition (Speed)

Speed is the distance travelled per unit of time. Writing d for distance, v for speed, and t for time,

$$v = \frac{d}{t}, \quad d = vt, \quad t = \frac{d}{v}.$$

These are three ways of writing one fact, not three facts. Given any two of d , v , t , you can produce the third.

Example 41

A cyclist covers 45 kilometres in 2 hours and 30 minutes. What is her speed?

Solution. First put the time in a single unit. Two hours and thirty minutes is

2.5 hours.

Now apply the definition:

$$v = \frac{d}{t} = \frac{45}{2.5} = 18$$

kilometres per hour.

Main idea/Fact (Label your quantities)

When a problem has more than one traveller, subscript everything: d_A , v_A , t_A for Andrea and d_L , v_L , t_L for Lauren.

Do this on paper while you are learning. In a contest you will often carry it in your head, but a mislabelled variable costs far more than the seconds the subscript takes.

Exercise 42. A train travels 30 miles in 45 minutes. What is its speed in miles per hour?⁴

The word “per” is a division bar

Every rate in a problem arrives disguised as an English phrase. Translating it is mechanical.

⁴Forty-five minutes is $\frac{3}{4}$ of an hour, so $v = 30 \div \frac{3}{4} = 40$ mph. It is not 30/45.

Main idea/Fact (Reading a rate) • “40 miles per hour” means $\frac{40 \text{ miles}}{1 \text{ hour}}$

- “30 miles per gallon” means $\frac{30 \text{ miles}}{1 \text{ gallon}}$
- “\$2.00 per gallon” means $\frac{2 \text{ dollars}}{1 \text{ gallon}}$
- “3 packages every 3 minutes” means $\frac{3 \text{ packages}}{3 \text{ minutes}} = \frac{1 \text{ package}}{1 \text{ minute}}$
- “a cupcake every 20 seconds” means $\frac{1}{20}$ cupcakes per second

The units of a rate tell you the formula. Whatever sits in the denominator plays the role that time plays in a speed problem.

Pitfall

The denominator is not always time. In “miles per gallon”, gallons occupy the denominator, so gallons behave exactly like hours: distance is the rate times the gallons, and gallons are the distance divided by the rate.

That is the whole reason the opening problem worked. Do not go looking for a time that the problem never mentions.

Pitfall

A rate and its reciprocal are different numbers with different units. “A cupcake every 20 seconds” is a rate of $\frac{1}{20}$ cupcakes per second, not 20. Confusing the two flips your final division and is the most common error in work problems.

Converting units

Rate problems mix minutes with hours and feet with miles constantly. Conversion is multiplication by a fraction equal to 1.

Main idea/Fact (Convert by multiplying by 1)

Since 1 mile is 5280 feet, the fraction

$$\frac{5280 \text{ feet}}{1 \text{ mile}}$$

equals 1, so you may multiply by it freely. Arrange the fraction so the unit you want to remove cancels.

Example 43

Convert 60 miles per hour into feet per second.

Solution. Start from the rate and multiply by conversion fractions, arranging each so the

unwanted unit cancels:

$$\frac{60 \text{ miles}}{1 \text{ hour}} \cdot \frac{5280 \text{ feet}}{1 \text{ mile}} \cdot \frac{1 \text{ hour}}{3600 \text{ seconds}}.$$

Miles cancel and hours cancel, leaving feet per second:

$$\frac{60 \cdot 5280}{3600} = \frac{316800}{3600} = 88$$

feet per second.

Exercise 44. A truck travels $\frac{b}{6}$ feet every t seconds. How many feet does it travel in 3 minutes?⁵

Splitting a trip into stretches

When the speed changes partway through, you cannot use one formula on the whole trip. Handle each stretch separately and then add the times.

Main idea/Fact (One stretch at a time)

Break the journey wherever the speed changes. Compute the time for each stretch with $t = \frac{d}{v}$, then add.

Times add. Speeds do not.

Example 45 (2021 Fall AMC 10A, Problem 4)

Mr. Lopez has a choice of two routes to get to work. Route A is 6 miles long, and his average speed along this route is 30 miles per hour. Route B is 5 miles long, and his average speed along this route is 40 miles per hour, except for a $\frac{1}{2}$ -mile stretch in a school zone where his average speed is 20 miles per hour. By how many minutes is Route B quicker than Route A?

(A) $2\frac{3}{4}$ (B) $3\frac{3}{4}$ (C) $4\frac{1}{2}$ (D) $5\frac{1}{2}$ (E) $6\frac{3}{4}$

Solution. The answer is asked for in minutes, so work in minutes throughout. A speed of 30 miles per hour is $\frac{30}{60} = \frac{1}{2}$ mile per minute, and 40 miles per hour is $\frac{2}{3}$ mile per minute, and 20 miles per hour is $\frac{1}{3}$ mile per minute.

Route A is a single stretch:

$$t_A = \frac{6}{1/2} = 12$$

minutes.

Route B splits into two stretches. The school zone is $\frac{1}{2}$ mile, so the rest is $5 - \frac{1}{2} = 4.5$ miles:

$$t_B = \frac{4.5}{2/3} + \frac{0.5}{1/3} = 6.75 + 1.5 = 8.25$$

minutes.

⁵Three minutes is 180 seconds, which is $180/t$ intervals, so the distance is $\frac{b}{6} \cdot \frac{180}{t} = \frac{30b}{t}$ feet. Converting to yards would divide by 3.

Therefore Route B saves

$$12 - 8.25 = 3.75 = \boxed{\text{(B)} 3\frac{3}{4}}$$

minutes.

Pitfall

Converting to minutes at the start was a choice, not a rule, but it paid off: the question asked for minutes, so no conversion was needed at the end. Decide your unit before you compute, based on what the final sentence wants.

Average speed is not the average of the speeds

This is the single most tested idea in this part.

Main idea/Fact (Average speed)

$$\text{average speed} = \frac{\text{total distance}}{\text{total time}}.$$

Always. Compute both totals and divide. Never average the speeds themselves.

Example 46

You drive 120 kilometres at 60 km/h, then return the same 120 kilometres at 40 km/h. What is your average speed for the round trip?

Solution. The outbound leg takes

$$\frac{120}{60} = 2$$

hours and the return leg takes

$$\frac{120}{40} = 3$$

hours. So the totals are 240 kilometres and 5 hours, giving

$$\frac{240}{5} = 48$$

kilometres per hour.

Pitfall

The average of 60 and 40 is 50, which is not the answer. You spend more time at the slower speed, so the slower speed counts for more. The true average is always below the plain average, and always closer to the slower speed.

Main idea/Fact (Equal distances at two speeds)

If the two halves of a trip cover the *same distance* at speeds v_1 and v_2 , the average speed is

$$\frac{2v_1v_2}{v_1 + v_2}.$$

Check it on the example: $\frac{2 \cdot 60 \cdot 40}{100} = 48$.

This formula also explains the opening problem: Ray and Tom drove equal distances at 40 and 10 miles per gallon, and

$$\frac{2 \cdot 40 \cdot 10}{50} = 16.$$

Learn the derivation, not the formula. The formula assumes equal distances, and problems frequently give equal *times* instead, in which case the plain average is correct.

Exercise 47. You ride 10 miles at 30 mph and another 10 miles at 60 mph. What is your average speed?⁶

Pick a convenient number

Many rate problems never tell you a distance, because the answer does not depend on it. When that happens, choose one.

Main idea/Fact (Choose the distance)

If the problem gives only speeds and ratios, set the unknown distance to a number that makes every division come out whole, usually the least common multiple of the speeds involved.

You may do this only when the answer genuinely does not depend on the choice. A ratio, a percentage, or an average speed is safe; an actual distance is not.

Example 48

A hiker climbs a hill at 3 km/h and descends the same path at 6 km/h. What is her average speed?

Solution. Let the path be 6 kilometres, chosen because 6 divides evenly by both 3 and 6. The climb takes 2 hours and the descent takes 1 hour, so

$$\frac{12}{3} = 4$$

kilometres per hour.

Had we chosen 12 kilometres instead, the times would have been 4 and 2, giving $\frac{24}{6} = 4$ again. The choice does not matter, which is exactly what licenses making it.

⁶Total distance 20 miles; total time $\frac{10}{30} + \frac{10}{60} = \frac{1}{2}$ hour; average 40 mph. Not 45.

Two movers: closing and chasing

When two objects move along the same line, stop tracking each one separately and track the **gap** between them instead.

Main idea/Fact (Relative speed) • **Toward each other:** the gap shrinks at the *sum* of the speeds, so

$$\text{time to meet} = \frac{\text{initial gap}}{v_1 + v_2}.$$

• **Same direction:** the gap shrinks at the *difference* of the speeds, so

$$\text{time to catch up} = \frac{\text{initial gap}}{v_{\text{fast}} - v_{\text{slow}}}.$$

Example 49 (2009 AMC 10A, Problem 20)

Andrea and Lauren are 20 kilometers apart. They bike toward one another with Andrea traveling three times as fast as Lauren, and the distance between them decreasing at a rate of 1 kilometer per minute. After 5 minutes, Andrea stops biking because of a flat tire and waits for Lauren. After how many minutes from the time they started to bike does Lauren reach Andrea?

(A) 20 (B) 30 (C) 55 (D) 65 (E) 80

Solution. They ride toward each other, so the gap closes at the sum of their speeds, and the problem hands us that sum directly:

$$v_A + v_L = 1$$

kilometre per minute. Since Andrea rides three times as fast, writing $v_L = v$ gives $v_A = 3v$ and

$$v + 3v = 1,$$

so $v = \frac{1}{4}$. Lauren rides at $\frac{1}{4}$ km/min.

Now split the trip into its two phases, because the relative speed formula only applies while both are moving. For the first 5 minutes the gap closes at 1 km/min, so it shrinks by 5 kilometres, leaving

$$20 - 5 = 15$$

kilometres. In the second phase only Lauren moves, so the relevant rate is her own $\frac{1}{4}$ km/min, and she needs

$$\frac{15}{1/4} = 60$$

minutes. The total is

$$5 + 60 = \boxed{\text{(D) } 65}$$

minutes.

Pitfall

The relative speed formulas apply only while *both* objects are moving steadily. A flat tire, a head start, or a change of road ends the phase.

Split the trip at every such moment, and use the sum or difference only on the stretch where both are actually moving.

Exercise 50. Two cars start 300 kilometres apart and drive toward each other at 60 and 40 km/h. How long until they meet?⁷

Meeting on a loop

This next one is a step up, and it is worth the effort because the idea recurs.

Main idea/Fact (Between two meetings on a loop)

When two runners go in opposite directions around a track, between any two consecutive meetings they together cover exactly one full lap.

So track the *combined* distance, not either runner's distance.

Example 51 (2004 AMC 10A, Problem 17)

Brenda and Sally run in opposite directions on a circular track, starting at diametrically opposite points. They first meet after Brenda has run 100 meters. They next meet after Sally has run 150 meters past their first meeting point. Each girl runs at a constant speed. What is the length of the track in meters?

(A) 250 (B) 300 (C) 350 (D) 400 (E) 500

Solution. Let the track length be x . They start half a lap apart, so up to the first meeting they together cover $\frac{x}{2}$. Between the first and second meetings they together cover a full lap, x .

The second stretch is twice the combined distance of the first, and their combined speed never changes, so the second stretch takes twice as long. Therefore each girl runs twice as far in it as she did before the first meeting. Brenda ran 100 metres before the first meeting, so she runs

$$2 \cdot 100 = 200$$

metres between the meetings.

Between the meetings Brenda and Sally together cover one full lap, so

$$x = 200 + 150 = \boxed{\text{(C) } 350}$$

metres.

⁷The gap closes at 100 km/h, so $300/100 = 3$ hours.

Work is a rate too

Nothing new is needed here. A work rate is jobs per unit of time, exactly as speed is distance per unit of time.

Definition (Work rate)

If a worker finishes a whole job in t units of time, that worker's rate is

$$\frac{1}{t}$$

jobs per unit of time. In general

$$\text{work} = \text{rate} \times \text{time}.$$

Main idea/Fact (Rates add)

When several workers work at the same time,

$$r_{\text{total}} = r_1 + r_2 + \cdots$$

Convert every time into a rate, add the rates, then convert back.

Example 52 (2012 AMC 10A, Problem 1)

Cagney can frost a cupcake every 20 seconds and Lacey can frost a cupcake every 30 seconds. Working together, how many cupcakes can they frost in 5 minutes?

(A) 10 (B) 15 (C) 20 (D) 25 (E) 30

Solution. Turn each time into a rate. Cagney frosts $\frac{1}{20}$ cupcakes per second and Lacey frosts $\frac{1}{30}$ cupcakes per second. Working together their rates add:

$$r = \frac{1}{20} + \frac{1}{30} = \frac{3}{60} + \frac{2}{60} = \frac{1}{12}$$

cupcakes per second, which is one cupcake every 12 seconds.

Five minutes is 300 seconds, so together they frost

$$\frac{1}{12} \cdot 300 = \boxed{\text{(D) } 25}$$

cupcakes.

Pitfall

Do not add or average the times. Two workers taking 20 and 30 seconds each do not take 50 seconds together, and they do not take 25 seconds either. They take 12.

Two workers together are always faster than the faster one alone, so the joint time must come out *below* the smaller of the two times. If it does not, you have flipped a reciprocal somewhere.

Main idea/Fact (Two workers, one job)

If each can finish the same job alone in x and y units of time, then together

$$r = \frac{1}{x} + \frac{1}{y} = \frac{x+y}{xy}, \quad \text{joint time} = \frac{xy}{x+y}.$$

Check it on the cupcakes: $\frac{20 \cdot 30}{50} = 12$.

Exercise 53. One tap fills a tub in 6 minutes and another fills it in 12 minutes. How long do they take together?⁸

Scaling a rate across two quantities

Sometimes a rate stretches over two variables at once. Reduce to a single unit of each, then scale back up.

Example 54 (2014 AMC 10A, Problem 6)

Suppose that a cows give b gallons of milk in c days. At this rate, how many gallons of milk will d cows give in e days?

- (A) $\frac{bde}{ac}$ (B) $\frac{ac}{bde}$ (C) $\frac{abde}{c}$ (D) $\frac{bcde}{a}$ (E) $\frac{abc}{de}$

Solution. Boil the given information down to one cow for one day.

In c days the a cows give b gallons, so in one day those a cows give $\frac{b}{c}$ gallons, so one cow gives

$$\frac{b}{ac}$$

gallons per day. That is the base rate.

Now scale up. With d cows for e days the milk is the base rate times the number of cows times the number of days:

$$\frac{b}{ac} \cdot d \cdot e = \boxed{\text{(A)} \frac{bde}{ac}}.$$

Main idea/Fact (All-letters problems)

When every quantity is a letter, substitute small friendly numbers, solve the concrete version, and see which answer choice matches.

Taking $a = b = c = d = e = 1$ collapses too many choices at once, so vary them: $a = 2$, $b = 6$, $c = 3$, $d = 4$, $e = 5$ separates the options quickly.

Sanity checks

Rate answers are easy to check, and the checks catch flipped reciprocals immediately.

⁸ $\frac{6 \cdot 12}{18} = 4$ minutes, which is indeed less than 6.

- Main idea/Fact (Before you bubble)**
- An average speed must lie strictly between the two speeds, and closer to the one held for longer.
 - Workers together must finish faster than the fastest one alone.
 - A time and a rate move in opposite directions: if your answer says a faster rate takes longer, you inverted something.
 - Reread the units of the answer choices. A question in minutes with an answer in hours is off by a factor of 60.

Summary

- Main idea/Fact (Setting up)**
- Identify the rate, and read its units to see what plays the role of time.
 - Write down which two of amount, rate, and time you know.
 - Convert everything to a single unit before computing, chosen to match what the question asks for.
 - Break the trip at every point where the speed changes, and handle each stretch separately.
 - If only speeds and ratios are given, choose a convenient distance, usually the least common multiple of the speeds.

- Main idea/Fact (Finishing)**
- Average speed is total distance over total time. Never average the speeds.
 - For two movers, track the gap: it closes at the sum of the speeds head-on and at the difference in a chase.
 - For work, convert times into rates, add the rates, then convert back. Times never add.
 - Check that the answer sits where it should: between the two speeds, or below the fastest single worker.

Problems

These are not sorted by difficulty. Try as many as you can before looking at the hints.

- (2013 AMC 10A #1) A taxi ride costs \$1.50 plus \$0.25 per mile traveled. How much does a 5-mile taxi ride cost?
(A) \$2.25 (B) \$2.50 (C) \$2.75 (D) \$3.00 (E) \$3.25
([Hint 1](#), [Hint 2](#))
- (2022 AMC 10A #2) Mike cycled 15 laps in 57 minutes. Assume he cycled at a constant speed throughout. Approximately how many laps did he complete in the first 27 minutes?
(A) 5 (B) 7 (C) 9 (D) 11 (E) 13
([Hint 1](#), [Hint 2](#))
- (2018 AMC 10B #2/12B #2) Sam drove 96 miles in 90 minutes. His average speed during the first 30 minutes was 60 mph (miles per hour), and his average speed during the second 30 minutes was 65 mph. What was his average speed, in mph, during the last 30 minutes?
(A) 64 (B) 65 (C) 66 (D) 67 (E) 68
([Hint 1](#), [Hint 2](#))
- (2020 AMC 10B #6) Driving along a highway, Megan noticed that her odometer showed 15951 (miles). This number is a palindrome: it reads the same forward and backward. Then 2 hours later, the odometer displayed the next higher palindrome. What was her average speed, in miles per hour, during this 2-hour period?
(A) 50 (B) 55 (C) 60 (D) 65 (E) 70
([Hint 1](#), [Hint 2](#))
- (2023 AMC 10A #1/12A #1) Cities A and B are 45 miles apart. Alicia lives in A and Beth lives in B . Alicia bikes towards B at 18 miles per hour. Leaving at the same time, Beth bikes toward A at 12 miles per hour. How many miles from City A will they be when they meet?
(A) 20 (B) 24 (C) 25 (D) 26 (E) 27
([Hint 1](#), [Hint 2](#))
- (2025 AMC 10A #1/12A #1) Andy and Betsy both live in Mathville. Andy leaves Mathville on his bicycle at 1:30, traveling due north at a steady 8 miles per hour. Betsy leaves on her bicycle from the same point at 2:30, traveling due east at a steady 12 miles per hour. At what time will they be exactly the same distance from their common starting point?
(A) 3:30 (B) 3:45 (C) 4:00 (D) 4:15 (E) 4:30
([Hint 1](#), [Hint 2](#))
- (2014 AMC 10B #8) A truck travels $\frac{b}{6}$ feet every t seconds. There are 3 feet in a yard. How many yards does the truck travel in 3 minutes?
(A) $\frac{b}{1080t}$ (B) $\frac{30t}{b}$ (C) $\frac{30b}{t}$ (D) $\frac{10t}{b}$ (E) $\frac{10b}{t}$
([Hint 1](#), [Hint 2](#))

8. (2012 AMC 10B #13) It takes Clea 60 seconds to walk down an escalator when it is not operating and only 24 seconds to walk down the escalator when it is operating. How many seconds does it take Clea to ride down the operating escalator when she just stands on it?

(A) 36 (B) 40 (C) 42 (D) 48 (E) 52

([Hint 1](#), [Hint 2](#))

9. (2020 AMC 10A #4/12A #3) A driver travels for 2 hours at 60 miles per hour, during which her car gets 30 miles per gallon of gasoline. She is paid \$0.50 per mile, and her only expense is gasoline at \$2.00 per gallon. What is her net rate of pay, in dollars per hour, after this expense?

(A) 20 (B) 22 (C) 24 (D) 25 (E) 26

([Hint 1](#), [Hint 2](#))

10. (2017 AMC 10B #7/12B #4) Samia set off on her bicycle to visit her friend, traveling at an average speed of 17 kilometers per hour. When she had gone half the distance to her friend's house, a tire went flat, and she walked the rest of the way at 5 kilometers per hour. In all it took her 44 minutes to reach her friend's house. In kilometers rounded to the nearest tenth, how far did Samia walk?

(A) 2.0 (B) 2.2 (C) 2.8 (D) 3.4 (E) 4.4

([Hint 1](#), [Hint 2](#))

11. (2011 AMC 10A #15) Roy bought a new battery-gasoline hybrid car. On a trip the car ran exclusively on its battery for the first 40 miles, then ran exclusively on gasoline for the rest of the trip, using gasoline at a rate of 0.02 gallons per mile. On the whole trip he averaged 55 miles per gallon. How long was the trip in miles?

(A) 140 (B) 240 (C) 440 (D) 640 (E) 840

([Hint 1](#), [Hint 2](#))

12. (2024 AMC 10A #8) Amy, Bomani, Charlie, and Daria work in a chocolate factory. On Monday Amy, Bomani, and Charlie started working at 1:00 PM and were able to pack 4, 3, and 3 packages, respectively, every 3 minutes. At some later time, Daria joined the group, and Daria was able to pack 5 packages every 4 minutes. Together, they finished packing 450 packages at exactly 2:45 PM. At what time did Daria join the group?

(A) 1:25 PM (B) 1:35 PM (C) 1:45 PM (D) 1:55 PM (E) 2:05 PM

([Hint 1](#), [Hint 2](#))

13. (2021 Spring AMC 10A #6) Chantal and Jean start hiking from a trailhead toward a fire tower. Jean is wearing a heavy backpack and walks slower. Chantal starts walking at 4 miles per hour. Halfway to the tower, the trail becomes really steep, and Chantal slows down to 2 miles per hour. After reaching the tower, she immediately turns around and descends the steep part of the trail at 3 miles per hour. She meets Jean at the halfway point. What was Jean's average speed, in miles per hour, until they meet?

(A) $\frac{12}{13}$ (B) 1 (C) $\frac{13}{12}$ (D) $\frac{24}{13}$ (E) 2

([Hint 1](#), [Hint 2](#))

Hints

Hint 1

1. **MAIN IDEA:** Only part of this cost is a rate. The \$1.50 is charged once no matter how far you go, so which piece of the cost gets multiplied by 5?

2. **MAIN IDEA:** Constant speed means laps and minutes are proportional. How many laps does Mike complete in one minute?

3. **MAIN IDEA:** You know the total distance, so find the distance covered in the first two stretches and subtract. How far does he go in 30 minutes at 60 mph? Careful: the speeds are per hour and the stretches are half-hours.

4. **MAIN IDEA:** The distance travelled is the difference between two odometer readings, so first find the next palindrome after 15951. Try increasing the middle digit and see what the palindrome condition forces.

5. **MAIN IDEA:** They ride toward each other, so track the gap between them. At what rate does the 45-mile gap shrink?

6. **MAIN IDEA:** The directions are a distraction: each one's distance from the start is just speed times their own riding time. The catch is that Andy has been riding an hour longer. If Betsy has ridden for t hours, how long has Andy ridden?

7. **MAIN IDEA:** Convert first, then count. Three minutes is how many seconds, and therefore how many t -second intervals?

8. **MAIN IDEA:** Two things move her down: her own walking and the escalator, and their rates add. Call the escalator's length d and write Clea's walking rate as a fraction of d per second.

9. **MAIN IDEA:** Net pay is money earned minus money spent, and the question wants it per hour. Everything flows from one number: how many miles does she drive in total?

10. **MAIN IDEA:** She covers the same distance twice, once by bike and once on foot, so name that half-distance d and write each time in terms of d . What do those two times have to add up to, and in what unit?

11. **MAIN IDEA:** The average of 55 miles per gallon is over the *whole* trip, but gasoline was only burned on part of it. If the trip is x miles long, how many of those miles used gasoline?

12. **MAIN IDEA:** Turn every “packages every few minutes” into packages per minute so the rates can be added. What is the combined rate of the first three workers, and what is Daria’s?

13. **MAIN IDEA:** No distance is given and none is needed, so choose one. Pick a half-trail length divisible by 4, 2, and 3, then compute how long Chantal is walking before the meeting.

Hint 2

1. The per-mile part contributes $5 \cdot 0.25$ dollars. Add the flat \$1.50 once, at the end.

2. His rate is $\frac{15}{57}$ laps per minute, so in 27 minutes he completes $\frac{15}{57} \cdot 27$ laps. The word “approximately” means you want the nearest listed value.

3. Half an hour at 60 mph is 30 miles and half an hour at 65 mph is 32.5 miles, for 62.5 miles. Subtract from 96 to get the last stretch, then divide by half an hour.

4. The next palindrome is 16061, so she drove $16061 - 15951$ miles in 2 hours. Divide.

5. The gap closes at $18 + 12 = 30$ mph, so they meet after $\frac{45}{30}$ hours. The question asks how far from A , so multiply that time by Alicia's speed, not Beth's.

6. If Betsy has ridden t hours then Andy has ridden $t + 1$ hours, so set $12t = 8(t + 1)$. Solve for t , then add that many hours to 2:30.

7. There are $\frac{180}{t}$ intervals of t seconds, each covering $\frac{b}{6}$ feet, for $\frac{30b}{t}$ feet. Now convert feet to yards, and check your answer against a small numerical case such as $b = 6$, $t = 1$.

8. Walking alone is $\frac{d}{60}$ per second and walking on the moving escalator is $\frac{d}{24}$ per second. The escalator's own rate is the difference. Then the standing time is d divided by that rate.

9. She drives 120 miles, earning $120 \cdot 0.50$ dollars and burning $\frac{120}{30}$ gallons at \$2.00 each. Subtract, then divide by 2 hours.

10. The times are $\frac{d}{17}$ and $\frac{d}{5}$ hours, and 44 minutes is $\frac{44}{60}$ of an hour, so

$$\frac{d}{17} + \frac{d}{5} = \frac{44}{60}.$$

Solve for d , and note that d is exactly the walking distance the question asks for.

11. If the trip is x miles, gasoline covers $x - 40$ miles and uses $0.02(x - 40)$ gallons. The whole-trip average gives

$$\frac{x}{0.02(x - 40)} = 55.$$

Solve for x .

12. The first three pack 10 packages every 3 minutes, so $\frac{10}{3}$ per minute over the full 105 minutes. Subtract what they packed from 450 to see how much Daria packed, then divide by her rate to find how long she worked, and count backwards from 2:45.

13. Taking the half-trail as 12 miles, Chantal needs 3, 6, and 4 hours for the three stretches, so 13 hours in total. In that same time Jean has walked 12 miles. Divide.

Percentages, Increase and Decrease

Introduction

Percentages are the last of the Week 0 algebra topics, and they are the one the AMC returns to most often. Here is what a hard version looks like.

Example 55 (2017 AMC 10B, Problem 11)

At Typico High School, 60% of the students like dancing, and the rest dislike it. Of those who like dancing, 80% say that they like it, and the rest say that they dislike it. Of those who dislike dancing, 90% say that they dislike it, and the rest say that they like it. What fraction of students who say they dislike dancing actually like it?

(A) 10% (B) 12% (C) 20% (D) 25% (E) $33\frac{1}{3}\%$

Solution. Every percent here is taken of a different group, so the way through is to stop working with percents and start counting students. Take 100 students.

Then 60 like dancing and 40 dislike it. Of the 60 who like it, 80% say so, which is 48, and the other 12 say they dislike it. Of the 40 who dislike it, 90% say so, which is 36, and the other 4 say they like it.

Now collect the students who *say* they dislike dancing:

$$12 + 36 = 48.$$

Of those 48, the ones who actually like dancing are the 12 from the first group. So the answer is

$$\frac{12}{48} = \boxed{\text{(D) } 25\%}.$$

Remark. A quarter of the students claiming to dislike dancing are lying about it, which is far more than most people guess. The reason is that the group who like dancing is larger to begin with, so even its small 20% minority is comparable in size to the honest majority of the smaller group.

Notice what made that tractable. We never added or averaged a single percent. We converted each one into a count of students, and only divided at the very end.

The main habit

There is really only one skill in this topic, and it is a translation skill. Every percent is a number waiting to become a decimal, and every percent is attached to some base.

Main idea/Fact (Main habit for percentages)

A percent is a number over 100. Turn every percent into a decimal, attach it to the right *base*, then do ordinary arithmetic.

Before you compute anything, answer one question out loud:

percent OF WHAT?

Almost every percentage mistake on the AMC is a wrong answer to that question.

Percent means “per hundred”

Definition (Percent)

Percent means “per hundred”. The symbol % is shorthand for $\frac{1}{100}$, so

$$x\% = \frac{x}{100}.$$

So $37\% = 0.37$ and $7\% = 0.07$. To take a percent *of* a quantity, the word “of” means multiply, exactly as it did in the word problems part.

Main idea/Fact (Percent of a number)

To take $x\%$ of a quantity N , compute

$$\frac{x}{100} \cdot N.$$

Example 56

What is 35% of 80?

Solution. The phrase translates directly:

$$\frac{35}{100} \cdot 80 = 0.35 \cdot 80 = 28.$$

That is the entire game: rewrite the percent as a decimal, multiply by the base.

Pitfall

A percent is never a number by itself. It is always a percent *of something*. When two quantities are floating around, write down which one the percent attaches to before you multiply.

Watch single-digit percents especially: 7% is 0.07, not 0.7.

Exercise 57. What is 7% of 250?⁹

Fractions worth knowing cold

Contest percentages are chosen so the arithmetic is clean in fraction form. Recognising these on sight saves real time.

Main idea/Fact (Convert on sight)

$$\begin{aligned} \frac{1}{2} &= 50\%, & \frac{1}{3} &= 33\frac{1}{3}\%, & \frac{2}{3} &= 66\frac{2}{3}\%, & \frac{1}{4} &= 25\%, & \frac{3}{4} &= 75\%, \\ \frac{1}{5} &= 20\%, & \frac{1}{6} &= 16\frac{2}{3}\%, & \frac{1}{8} &= 12.5\%, & \frac{3}{8} &= 37.5\%, & \frac{5}{8} &= 62.5\%, & \frac{7}{8} &= 87.5\%. \end{aligned}$$

When a problem hands you 37.5%, it is really handing you $\frac{3}{8}$.

⁹ $0.07 \cdot 250 = 17.5$. If you got 175, you dropped a decimal place.

Part, whole, and percent

Every basic percent problem is one equation seen from three sides.

Main idea/Fact (The percent relationship)

$$\text{part} = \text{rate} \times \text{whole}, \quad \text{rate} = \frac{x}{100}.$$

Whichever piece is missing, solve for it:

$$\text{part} = \text{rate} \cdot \text{whole}, \quad \text{rate} = \frac{\text{part}}{\text{whole}}, \quad \text{whole} = \frac{\text{part}}{\text{rate}}.$$

This is the same shape as $\text{amount} = \text{rate} \times \text{time}$ from the previous part, and for the same reason: a percent is a rate.

Example 58

Using the whole 80: (a) what is 35% of 80? (b) 28 is what percent of 80? (c) 28 is 35% of what number?

Solution. These are one fact rearranged three ways. For (a) the part is

$$0.35 \cdot 80 = 28.$$

For (b) the rate is

$$\frac{28}{80} = 0.35 = 35\%.$$

For (c) the whole is

$$\frac{28}{0.35} = 80.$$

That third version, recovering the whole from the part, is called a **reverse percent**, and it is where most points are lost.

Example 59

A jacket sells for \$48 after 20% off. What was the original price?

Solution. The \$48 is not the discount. It is what remains after 20% has been removed, so it is 80% of the original:

$$0.80 \cdot (\text{original}) = 48.$$

Undoing a multiplication means dividing:

$$\text{original} = \frac{48}{0.80} = 60.$$

The jacket was \$60.

Pitfall

For “20% off, sale price \$48”, do not take 20% of 48. The \$48 is a percent of the unknown original, so you must *divide* by the multiplier rather than multiply by it.

Check by running it forwards: 20% off \$60 is \$48. If running it forwards misses the given number, you multiplied where you should have divided.

Exercise 60. A shirt is on sale for \$60 after a 25% discount. What was its original price?¹⁰

Choose a convenient whole

Many percentage problems never name a total, because the answer does not depend on it. When that happens, pick one, and pick 100.

Main idea/Fact (Set the whole to 100)

If a problem gives only percentages and asks for a percentage, let the total be 100.

Every percent then becomes a count, the arithmetic becomes integer arithmetic, and the final answer is already a percent. Use a different number only when 100 divides badly, for instance when the problem involves sevenths.

That is exactly what carried the opening problem. It is also the same move as choosing a convenient distance in the speed part, and it is legitimate for the same reason: the answer does not depend on the choice.

Example 61

In a town, 60% of residents own a bicycle and 45% of residents own both a bicycle and a helmet. What percent of bicycle owners own a helmet?

Solution. Let the town have 100 residents, so 60 own a bicycle and 45 own both.

The question makes bicycle owners the whole, so the base is 60 rather than 100:

$$\frac{45}{60} = 0.75 = 75\%.$$

Pitfall

Notice how quietly the base changed in that last step. “What percent of bicycle owners” replaces the town by the 60 cyclists. Reread the final sentence and identify the new whole before dividing.

Percent of a percent

When one percent is taken of another, the rates multiply.

Main idea/Fact (Chaining percents)

Taking $q\%$ of $p\%$ of a whole gives

$$\frac{q}{100} \cdot \frac{p}{100} \cdot (\text{whole}).$$

The two percents multiply. They never add.

¹⁰The \$60 is 75% of the original, so the original is $60/0.75 = 80$ dollars.

Example 62

In a club, 60% of members are juniors, and 25% of the juniors play chess. What percent of the whole club are chess-playing juniors?

Solution. Take 100 members. Then 60 are juniors, and the chess players are 25% of those 60:

$$0.25 \cdot 60 = 15,$$

so 15% of the club. Equivalently $0.60 \cdot 0.25 = 0.15$ in one step.

Pitfall

25% of 60% is 15% of the whole. It is not 85% and not 35%. Adding or subtracting percents only makes sense when both are taken of the *same* base.

Percent increase and decrease

Main idea/Fact (The multiplier method) • Increasing by $p\%$ multiplies by $\left(1 + \frac{p}{100}\right)$.

- Decreasing by $p\%$ multiplies by $\left(1 - \frac{p}{100}\right)$.

And the percent change itself is

$$\text{percent change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100\%,$$

where the denominator is always the *old* value.

Example 63

An \$80 jacket is marked up 25%, then put on sale at 25% off the new price. What is the final price?

Solution. The mark-up multiplies by 1.25:

$$80 \cdot 1.25 = 100.$$

The sale then multiplies by 0.75:

$$100 \cdot 0.75 = 75.$$

The final price is \$75, not \$80. The increase and the decrease were taken of different bases, so they do not cancel.

Exercise 64. A town's population grows from 250 to 320. What is the percent increase?¹¹

Percentage points are not percents

This distinction is small, easy to state, and regularly exploited.

¹¹The change is 70 and the base is the old value 250, so $70/250 = 0.28$, a 28% increase.

Main idea/Fact (Two ways to compare percents)

If a rate moves from 40% to 45%, then it rose by

- 5 **percentage points**, which is the plain difference, and
- 12.5%, which is the percent change, since $\frac{45 - 40}{40} = 0.125$.

Pitfall

“The tax rate rose by 5%” and “the tax rate rose by 5 percentage points” mean different things. Read which one the problem states, and answer in the unit the question asks for.

Exercise 65. A sales tax rises from 20% to 25%. By how many percentage points did it rise, and by what percent did it rise?¹²

“More than” does not reverse symmetrically**Main idea/Fact** (Flipping the comparison)

If A is $p\%$ more than B , then

$$A = \left(1 + \frac{p}{100}\right) B, \quad \text{so} \quad B = \frac{A}{1 + \frac{p}{100}}.$$

The percent by which B is less than A is *not* p . Recompute it from the new base A .

Example 66

Sam earns 25% more than Tara. By what percent is Tara’s salary less than Sam’s?

Solution. Let Tara earn 100, so Sam earns 125. The question asks how much less Tara is than Sam, which makes Sam the base:

$$\frac{125 - 100}{125} = \frac{25}{125} = 0.20,$$

so Tara earns 20% less. The numbers 25 and 20 differ because they are percentages of different quantities.

Exercise 67. If A is 50% more than B , by what percent is B less than A ?¹³

Stacking changes: multipliers multiply

¹²Five percentage points, and $5/20 = 25\%$.

¹³Take $B = 100$, so $A = 150$, and $\frac{50}{150} = 33\frac{1}{3}\%$.

Main idea/Fact (Successive changes multiply)

A change of $p\%$ followed by a change of $q\%$ replaces the original by

$$(1 \pm \frac{p}{100}) (1 \pm \frac{q}{100}) \times (\text{original}).$$

You never add the percents. You multiply the multipliers.

The order does not matter, since multiplication commutes.

Example 68 (2019 AMC 8, Problem 22)

A store increased the original price of a shirt by a certain percent and then decreased the new price by the same amount. Given that the resulting price was 84% of the original price, by what percent was the price increased and decreased?

(A) 16 (B) 20 (C) 28 (D) 36 (E) 40

Solution. Up by $p\%$ and then down by $p\%$ multiplies by

$$(1 + \frac{p}{100}) (1 - \frac{p}{100}),$$

and by the difference of squares this is

$$1 - (\frac{p}{100})^2 = 0.84.$$

So

$$(\frac{p}{100})^2 = 0.16,$$

giving $\frac{p}{100} = 0.4$ and

$$p = \boxed{\text{(E) } 40}.$$

Pitfall

A $p\%$ increase followed by a $p\%$ decrease never returns you to the start. The decrease is taken of a *larger* base than the increase was, so the result is always $1 - (\frac{p}{100})^2$ of the original, which is below 1.

Exercise 69. A stock rises 20% one day and falls 20% the next. What percent of its original value remains?¹⁴

Exercise 70. An item's price is reduced by 10%, then a 10% sales tax is applied to the reduced price. The final cost is \$59.40. What was the original price?¹⁵

Blended percentages

When two groups merge, the overall percent is a *weighted average*, weighted by the sizes of the groups.

¹⁴ $1.20 \cdot 0.80 = 0.96$, so 96% remains. The stock is down, not even.

¹⁵The combined multiplier is $0.90 \cdot 1.10 = 0.99$, so the original is $59.40/0.99 = 60$ dollars.

Main idea/Fact (Weight by the base)

When amounts A and B are combined,

$$\text{overall rate} = \frac{(\text{rate of } A) \cdot A + (\text{rate of } B) \cdot B}{A + B}.$$

The denominator is the *combined* base. Track the actual amounts, then divide at the end.

Example 71 (2013 AMC 10A, Problem 10)

A flower bouquet contains pink roses, red roses, pink carnations, and red carnations. One third of the pink flowers are roses, three fourths of the red flowers are carnations, and six tenths of the flowers are pink. What percent of the flowers are carnations?

(A) 15 (B) 30 (C) 40 (D) 60 (E) 70

Solution. Every flower carries two labels at once, a colour and a type, so split by colour first and take 100 flowers. Six tenths are pink, so there are 60 pink and 40 red.

Among the pink flowers, one third are roses, so two thirds are carnations:

$$\frac{2}{3} \cdot 60 = 40.$$

Among the red flowers, three fourths are carnations:

$$\frac{3}{4} \cdot 40 = 30.$$

Altogether there are $40 + 30 = 70$ carnations out of 100 flowers, so

(E) 70

percent.

Remark. The two carnation rates were $66\frac{2}{3}\%$ among pink flowers and 75% among red, and the overall answer of 70% sits between them, nearer the pink rate because the pink group is larger. That is what a weighted average always looks like.

Pitfall

Two traps live in blended problems. First, never average the two rates directly, since each must be weighted by its group size. Second, read which side of the split is described, roses versus carnations, or “like” versus “say they like”, and answer the side actually asked for.

Exercise 72. A chemist mixes 40 mL of a 10% saline solution with 60 mL of a 30% saline solution. What is the concentration of the mixture?¹⁶

Backing out one group

The weighted average runs in reverse too. Knowing the overall rate and one group’s rate, you can recover the other.

¹⁶Salt: $4 + 18 = 22$ mL in 100 mL, so 22%. Not 20%, since more of the stronger solution was used.

Main idea/Fact (Work in totals, not rates)

Convert each average into a total, subtract the group you know, then divide by the size of the group you want.

Example 73

A class of 25 students averaged 78% on a test. The 10 boys averaged 75%. What did the 15 girls average?

Solution. The class accounts for

$$78 \cdot 25 = 1950$$

percentage points in total, and the boys account for

$$75 \cdot 10 = 750.$$

So the girls account for $1950 - 750 = 1200$ points among 15 students:

$$\frac{1200}{15} = 80\%.$$

Exercise 74. A group of 20 people has an average age of 30. The 8 children average 10. What is the adults' average age?¹⁷

Sanity checks

Main idea/Fact (Before you bubble) • A blended rate must lie between the two group rates, nearer the larger group.

- After a discount, the price must go down; after a tax, up. If your multiplier points the wrong way, you added when you should have subtracted.
- A reverse percent answer should reproduce the given number when you run the change forwards.
- Reread the final sentence. Problems routinely ask for the percent *off* rather than the percent remaining, or for the group you did not solve for.

Summary

Main idea/Fact (Setting up) • Write $x\% = \frac{x}{100}$, then ask **percent OF WHAT?** and attach the rate to that base.

- If the problem gives only percentages, set the whole to 100 and count.
- $\text{part} = \text{rate} \times \text{whole}$. Missing the whole means you *divide*.
- Percents of the same base may be added. Percents of different bases may not.

¹⁷Total 600, children 80, so the adults total 520 among 12 people, giving $43\frac{1}{3}$.

Main idea/Fact (Finishing) • Increase by $p\%$ multiplies by $1 + \frac{p}{100}$, decrease multiplies by $1 - \frac{p}{100}$, and percent change uses the *old* value as denominator.

- Stacked changes multiply their multipliers. An equal up-then-down always lands below the start.
- “ $p\%$ more” does not reverse into “ $p\%$ less”. Recompute from the new base.
- Blending groups is a weighted average by group size. Track real amounts and divide at the end.
- Percentage points and percents are different units. Answer in the one asked for.

Problems

These are not sorted by difficulty. Try as many as you can before looking at the hints.

1. (2010 AMC 10B #2/12B #1) Makarla attended two meetings during her 9-hour work day. The first meeting took 45 minutes and the second meeting took twice as long. What percent of her work day was spent attending meetings?

(A) 15 (B) 20 (C) 25 (D) 30 (E) 35

([Hint 1](#), [Hint 2](#))

2. (2010 AMC 8 #9) Ryan got 80% of the problems correct on a 25-problem test, 90% on a 40-problem test, and 70% on a 10-problem test. What percent of all the problems did Ryan answer correctly?

(A) 63 (B) 75 (C) 80 (D) 84 (E) 86

([Hint 1](#), [Hint 2](#))

3. (2012 AMC 8 #8) A shop advertises that everything is “half price in today’s sale.” In addition, a coupon gives a 20% discount on sale prices. Using the coupon, the price today represents what percentage discount off the original price?

(A) 10 (B) 33 (C) 40 (D) 60 (E) 70

([Hint 1](#), [Hint 2](#))

4. (2013 AMC 8 #12) At the 2013 Winnebago County Fair a vendor is offering a “fair special” on sandals. If you buy one pair of sandals at the regular price of \$50, you get a second pair at a 40% discount, and a third pair at half the regular price. Javier took advantage of the “fair special” to buy three pairs of sandals. What percentage of the \$150 regular price did he save?

(A) 25 (B) 30 (C) 33 (D) 40 (E) 45

([Hint 1](#), [Hint 2](#))

5. (2019 AMC 12A #1) The area of a pizza with radius 4 inches is N percent larger than the area of a pizza with radius 3 inches. What is the integer closest to N ?

(A) 25 (B) 33 (C) 44 (D) 66 (E) 78

([Hint 1](#), [Hint 2](#))

6. (2019 AMC 12A #2) Suppose a is 150% of b . What percent of a is $3b$?

(A) 50 (B) $66\frac{2}{3}$ (C) 150 (D) 200 (E) 450

([Hint 1](#), [Hint 2](#))

7. (2018 AMC 10A #2) Liliane has 50% more soda than Jacqueline, and Alice has 25% more soda than Jacqueline. What is the relationship between the amounts of soda that Liliane and Alice have?

(A) Liliane has 20% more soda than Alice.

(B) Liliane has 25% more soda than Alice.

(C) Liliane has 45% more soda than Alice.

(D) Liliane has 75% more soda than Alice.

(E) Liliane has 100% more soda than Alice.

([Hint 1](#), [Hint 2](#))

8. (2011 AMC 10A #8) Last summer 30% of the birds living on Town Lake were geese, 25% were swans, 10% were herons, and 35% were ducks. What percent of the birds that were not swans were geese?

(A) 20 (B) 30 (C) 40 (D) 50 (E) 60

([Hint 1](#), [Hint 2](#))

9. (2014 AMC 10B #7) Suppose $A > B > 0$ and A is $x\%$ greater than B . What is x ?

(A) $100\left(\frac{A-B}{B}\right)$ (B) $100\left(\frac{A+B}{B}\right)$ (C) $100\left(\frac{A+B}{A}\right)$ (D) $100\left(\frac{A-B}{A}\right)$ (E) $100\left(\frac{A}{B}\right)$

([Hint 1](#), [Hint 2](#))

10. (2013 AMC 10A #9) In a recent basketball game, Shenille attempted only three-point shots and two-point shots. She was successful on 20% of her three-point shots and 30% of her two-point shots. Shenille attempted 30 shots. How many points did she score?

(A) 12 (B) 18 (C) 24 (D) 30 (E) 36

([Hint 1](#), [Hint 2](#))

11. (2017 AMC 8 #14) Chloe and Zoe are both students in Ms. Demeanor's math class. Last night they each solved half of the problems in their homework assignment alone and then solved the other half together. Chloe had correct answers to only 80% of the problems she solved alone, but overall 88% of her answers were correct. Zoe had correct answers to 90% of the problems she solved alone. What was Zoe's overall percentage of correct answers?

(A) 89 (B) 92 (C) 93 (D) 96 (E) 98

([Hint 1](#), [Hint 2](#))

12. (2015 AMC 10A #9) Two right circular cylinders have the same volume. The radius of the second cylinder is 10% more than the radius of the first. What is the relationship between the heights of the two cylinders?

(A) The second height is 10% less than the first.

(B) The first height is 10% more than the second.

(C) The second height is 21% less than the first.

(D) The first height is 21% more than the second.

(E) The second height is 80% of the first.

([Hint 1](#), [Hint 2](#))

13. (2017 AMC 10B #12) Elmer's new car gives 50% percent better fuel efficiency, measured in kilometers per liter, than his old car. However, his new car uses diesel fuel, which is 20% more expensive per liter than the gasoline his old car used. By what percent will Elmer save money if he uses his new car instead of his old car for a long trip?

(A) 20% (B) $26\frac{2}{3}\%$ (C) $27\frac{7}{9}\%$ (D) $33\frac{1}{3}\%$ (E) $66\frac{2}{3}\%$

[\(Hint 1, Hint 2\)](#)

14. (2014 AMC 10B #11) For the consumer, a single discount of $n\%$ is more advantageous than any of the following discounts:

- (1) two successive 15% discounts
- (2) three successive 10% discounts
- (3) a 25% discount followed by a 5% discount

What is the smallest possible positive integer value of n ?

- (A) 27 (B) 28 (C) 29 (D) 31 (E) 33

[\(Hint 1, Hint 2\)](#)

15. (2021 Fall AMC 10B #9) The knights in a certain kingdom come in two colors. $\frac{2}{7}$ of them are red, and the rest are blue. Furthermore, $\frac{1}{6}$ of the knights are magical, and the fraction of red knights who are magical is 2 times the fraction of blue knights who are magical. What fraction of red knights are magical?

- (A) $\frac{2}{9}$ (B) $\frac{3}{13}$ (C) $\frac{7}{27}$ (D) $\frac{2}{7}$ (E) $\frac{1}{3}$

[\(Hint 1, Hint 2\)](#)

Hints

Hint 1

1. **MAIN IDEA:** The percent is part over whole, and both must be in the same unit before you divide. Total the meeting time in minutes, then convert the 9-hour day into minutes too.

How many minutes do the two meetings take altogether?

2. **MAIN IDEA:** The overall percent is a weighted average, since each test counts in proportion to how many problems it holds. Averaging the three percentages outright will mislead you, so convert each into an actual count of correct answers.

How many problems did Ryan get right on each test?

3. **MAIN IDEA:** Two discounts in a row act on different bases, so adding 50 and 20 will mislead you. Set the original price to \$100 and apply the discounts one at a time.

What is the price after the half-price cut?

4. **MAIN IDEA:** “Percent saved” compares the saving against the full \$150, so start by finding what Javier actually pays under the special.

What does each of the three pairs cost?

5. **MAIN IDEA:** “Percent larger” compares the difference against the smaller quantity, and the quantities compared here are areas rather than radii. Write both areas as πr^2 .

What are the two areas, and does the π survive the comparison?

6. **MAIN IDEA:** Translate the sentence before anything else, since “ a is 150% of b ” says $a = 1.5b$. Then read the question closely, because “what percent *of* a ” makes a the whole.

Which ratio do you need to compute?

7. **MAIN IDEA:** Both given percents are taken of Jacqueline, so make her the base. Give her 100 units of soda and turn each “more than” into a multiplier.

How much do Liliane and Alice each have then?

8. **MAIN IDEA:** The question asks for geese as a percent of the birds that are *not* swans, so the whole is no longer all the birds. Take 100 birds and find the non-swan total first.

How many of the 100 birds are not swans?

9. **MAIN IDEA:** Translate “ A is $x\%$ greater than B ” into an equation using the multiplier form, then solve that equation for x . If the algebra feels slippery, test with $B = 100$ and $A = 120$ and see which choice returns 20.

What equation does the sentence become?

10. **MAIN IDEA:** You are not told how many of each kind of shot she attempted, so name it. Let x be the number of three-point attempts and write everything else in terms of x .

How many points come from the three-point shots, and how many from the two-point shots?

11. **MAIN IDEA:** Both students faced the same two halves, a solo half and a shared half, and they solved the shared half together, so both scored the same on it. Chloe's data pins down that shared score.

From Chloe's numbers alone, what percent of the shared half was correct?

12. **MAIN IDEA:** Equal volumes means $\pi r_1^2 h_1 = \pi r_2^2 h_2$, and the radius enters squared, so a 10% change in radius is not a 10% effect on height. Write $r_2 = 1.1r_1$ and solve for the ratio of the heights.

What is h_1/h_2 ?

13. **MAIN IDEA:** Two independent rates interact: one changes how many liters the trip needs, the other changes the price per liter. Set the old car to 1 km per liter and the old fuel to \$1 per liter, then compute the cost per kilometer for each car.

What does the new car cost per kilometer?

14. **MAIN IDEA:** Convert each package of discounts into the single multiplier it applies to the price, remembering that successive discounts multiply, so two 15% discounts leave $(0.85)^2$.

Which of the three packages leaves the customer paying the least, and which the most?

15. **MAIN IDEA:** This is the blended average written in fractions. The magical knights split into red magical and blue magical, and those two counts must together make $\frac{1}{6}$ of all knights.

If the fraction of blue knights who are magical is f , what is the fraction for red knights?

Hint 2

1. The meetings take $45 + 2 \cdot 45 = 135$ minutes out of a $9 \cdot 60 = 540$ minute day, so the rate is $\frac{135}{540}$.

What percent is that fraction?

2. He answered $0.80 \cdot 25 = 20$, then $0.90 \cdot 40 = 36$, then $0.70 \cdot 10 = 7$ problems correctly, out of $25 + 40 + 10 = 75$ in all.

What percent is $\frac{63}{75}$?

3. After the half-price cut the coupon removes 20% of *that* reduced price, so it multiplies by 0.80, and the shopper pays $0.50 \cdot 0.80$ of the original. But the question asks the percent taken *off*, not the percent remaining.

What percent has been removed?

4. The three pairs cost $50 + 0.60 \cdot 50 + 0.50 \cdot 50$, so subtract that total from 150 to get the saving, then divide the saving by 150.

What percent does that give?

5. The comparison is

$$\frac{16\pi - 9\pi}{9\pi} = \frac{7}{9},$$

with the π cancelling. Convert $\frac{7}{9}$ into a percent.

Which integer is that closest to?

6. The ratio is

$$\frac{3b}{a} = \frac{3b}{1.5b},$$

and the b cancels completely. Convert what remains into a percent.

What percent is that?

7. Liliane holds $1.50 \cdot 100 = 150$ and Alice holds $1.25 \cdot 100 = 125$. The answer choices compare Liliane to *Alice*, so Alice becomes the new base: compute $\frac{150-125}{125}$.

What percent more is that?

8. With 100 birds, 25 are swans, so 75 are not, and 30 of those 75 are geese. The answer is $\frac{30}{75}$ as a percent.

What does that simplify to?

9. The sentence says $A = B \left(1 + \frac{x}{100}\right)$, so

$$\frac{x}{100} = \frac{A - B}{B}.$$

Multiply through by 100 to isolate x .

Which choice matches, and does it return 20 when $B = 100$ and $A = 120$?

10. With x three-point attempts she makes $0.20x$ of them, worth $3(0.20x)$ points, and with $30 - x$ two-point attempts she makes $0.30(30 - x)$ of them, worth $2(0.30)(30 - x)$ points. Add the two expressions and simplify.

What happens to the x terms, and what does that tell you?

11. Take 100 problems, so each half has 50. Chloe got $0.80 \cdot 50 = 40$ right alone and $0.88 \cdot 100 = 88$ right overall, leaving 48 correct out of the 50 shared problems. Zoe got $0.90 \cdot 50 = 45$ right alone and the same 48 on the shared half.

Adding Zoe's two totals and dividing by 100, what is her overall percentage?

12. Cancelling π from $r_1^2 h_1 = r_2^2 h_2$ and substituting $r_2 = 1.1r_1$ gives $h_1 = 1.21h_2$. Now read the choices carefully, since “the first is $p\%$ more than the second” and “the second is $p\%$ less than the first” are different statements.

Which of those two forms does $h_1 = 1.21h_2$ say directly?

13. The old car costs \$1 per kilometer. The new car covers 1.5 km per liter on fuel costing \$1.20 per liter, so it costs $\frac{1.20}{1.5}$ dollars per kilometer.

What is that cost, and what percent below \$1 does it sit?

14. The three packages leave $(0.85)^2 = 0.7225$, $(0.9)^3 = 0.729$, and $0.75 \cdot 0.95 = 0.7125$ of the price. To beat all three, a single $n\%$ discount must leave the customer paying less than the smallest of those.

What is the smallest integer n with $1 - \frac{n}{100} < 0.7125$?

15. Writing f for the blue fraction and $2f$ for the red fraction, the magical knights satisfy

$$\frac{2}{7}(2f) + \frac{5}{7}(f) = \frac{1}{6}.$$

Solve for f , then remember the question asks for the *red* fraction.

What is $2f$?

Shef Scholars

Ensuring no Ramanujan or Mirzakhani is ever missed again

